Experimental assessment of a gas-conserving one-dimensional framework for entrapped-air-driven geysering

Zijian Xue and David Z. Zhu*

Abstract

Entrapped-air release through water-filled shafts can cause destructive geysering, but network-scale models often prescribe gas-release histories and cannot predict venting-to-geysering transitions. We evaluate gas-conserving one-dimensional mixed-flow realizations against three published campaigns. Phase inventories are conserved, release histories and geyser heights are not imposed, and settings are fixed within each series apart from one disclosed in-sample screen. Ventilation-shaft outcomes are reproduced despite an early, high narrow-shaft interface. The horizontal-pipe/vertical-riser realization reproduces 5 of 7 filmed outcomes and 39 of 63 criterion classifications; all 24 disagreements are missed geysers. Junction-chamber results capture the A2/B3 outcome reversal and first C9 pressure peak but underpredict the B3 peak. The framework supports system-scale analysis, but its one-sided bias precludes stand-alone hazard screening near the regime boundary.

Keywords: geysering, entrapped air pocket, stormwater system, vertical shaft, two-fluid model, transient mixed flow

*Corresponding author

1. Introduction

Geysering is the rapid expulsion of an air–water mixture through a manhole, dropshaft, or ventilation shaft during transient filling of a drainage system. The resulting jet can displace covers, damage nearby infrastructure, and endanger the public. Field pressure records rule out a purely inertial water-column explanation because the piezometric head near an erupting shaft need not reach ground level [1]. Controlled experiments instead identify the release and expansion of a large entrapped air pocket through a partly or fully water-filled shaft as the principal driving mechanism [1, 2].

Laboratory studies have documented several manifestations of this mechanism. Vasconcelos and Wright [2] isolated air-pocket release in a single ventilation shaft and documented both benign venting and geysering. Cong et al. [3] varied riser diameter and initial pocket volume systematically and proposed a two-parameter regime criterion. Later studies examined larger shafts [4], prototype-motivated junction chambers [5], covered manholes [6], and mitigation measures [7]. Related multiphase-flow work has also isolated the rapid expansion of a Taylor bubble in a vertical liquid column [8]. These studies indicate that pressure alone does not determine geysering. The outcome depends on pocket compression, gas delivery to the shaft, counter-current exchange, and the time available for the water column to reach the rim.

Numerical approaches resolve this competition at different levels of detail. Three-dimensional volume-of-fluid simulations can reproduce the local interface and jet structure of an individual event [9]. Their computational cost limits application to network-scale studies with many shafts and rainfall scenarios. One-dimensional mixed-flow models are more practical at that scale. Conventional Preissmann-slot and two-component-pressure formulations capture transitions between free-surface and pressurized flow but do not, by themselves, conserve and transport a distinct gas phase [10]. Entrapped gas is therefore commonly introduced through a lumped pocket law, a specified release rate, or a boundary condition. Rigid-column two-phase models likewise require the gas-pocket and water-column topology to be known in advance [2]. These approaches can represent a selected apparatus once the gas topology or release has been specified. They cannot independently determine whether the computed gas–liquid dynamics produce venting or geysering.

A companion methods paper [11] addressed this modelling gap for transient mixed flow in closed conduits. The stratified branch conserves gas mass and momentum within a decoupled two-fluid system, and the liquid-full branch uses an elastic-pipe closure. Moving pressurized–stratified interfaces are treated as regime-transition Riemann problems and tracked by a conservative cut-cell update. The restoring coefficient also retains interfacial-shear slip, so loss of hyperbolicity identifies the onset of inviscid Kelvin–Helmholtz instability. This is an onset diagnostic rather than a closure for subsequent wave breaking or gas entrainment. The methods paper verified the numerical formulation with canonical mixed-flow tests and included a schematic geyser-onset example, but did not assess the model quantitatively against geysering experiments.

Here, we assess case-specific one-dimensional realizations derived from the companion framework against geysering experiments. The realizations share conservative phase inventories and connected-pocket accounting, while their regular-face fluxes, shaft reductions, and junction closures follow the needs of each apparatus. Their parameters and numerical settings are fixed before comparison, apart from the disclosed in-sample Test B screen. The C9 phase-one all-water variant is identified separately because it follows the analytic comparison in the source study. Gas release, water-column displacement, and regime selection are therefore computed outcomes. No measured release history or geyser trajectory is imposed.

The assessment draws on three complementary evidence sets. The first campaign compares time-resolved pressure, interface, and free-surface histories for the non-geysering and geysering ventilation-shaft tests of Vasconcelos and Wright [2] (Fig. 1). The second treats the model as a regime classifier against the seven filmed riser tests of Cong et al. [3]; a separate 63-configuration simulation sweep is compared with their empirical regime criterion. The third examines the junction-chamber experiments of Liu et al. [5], including a change in geysering outcome caused by the downstream boundary and a transient mass-oscillation event. Together, the campaigns examine whether the model reproduces the observed outcome, whether that outcome pattern is consistent with an empirical regime map, and whether the reduced representation predicts the initial response of a prototype-motivated chamber.

This study therefore examines when gas-conserving one-dimensional realizations reproduce observed geysering. It also identifies when unresolved junction, liquid-film, and pressure-wave processes control the error. Sections 2 and 3 summarize the governing formulation and numerical method,

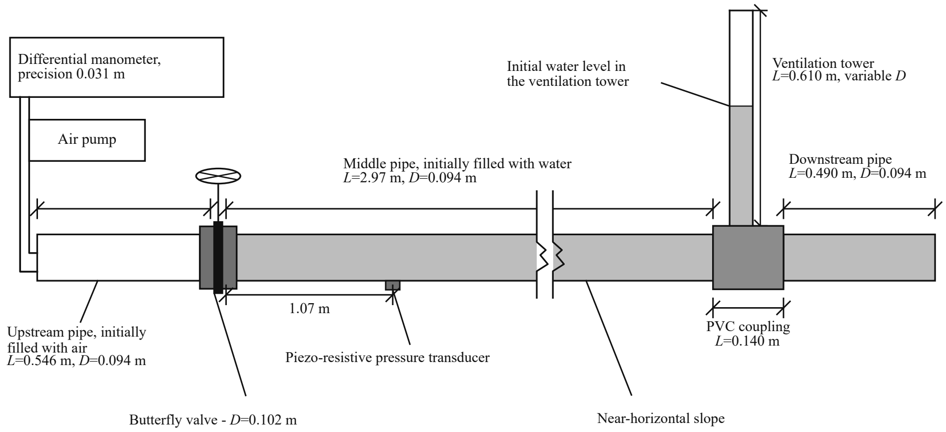

Figure 1: Campaign 1 apparatus, redrawn from Fig. 2 of [2].

including the application-specific closures. Section 4 reports the three experimental comparisons. Section 5 interprets the common successes and limitations, and Section 6 states the resulting scope of use.

2. Governing model

The governing equations below and the numerical discretization in Section 3 are taken directly from the companion methods manuscript submitted in parallel [11]. The same state variables, closures, regime limits, moving-front conditions, and conservative update are retained. The derivations, nonlinear-solver details, boundary catalogue, and canonical verification suite are omitted.

2.1. Decoupled two-fluid formulation

For a stratified gas–liquid section, the cross-sectionally averaged model is written

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S},$$

$$\mathbf{U} = \begin{bmatrix} A_g\rho_g \\ \rho_g Q_g \\ A_l \\ Q_l \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \rho_g Q_g \\ \rho_g Q_g^2/A_g + P_g A_g \\ Q_l \\ Q_l^2/A_l + \frac{1}{2}\Lambda_d A_l^2 \end{bmatrix}, \tag{1}$$

$$\mathbf{S} = \begin{bmatrix} 0 \\ P_g\,\partial_x A_g - A_g\rho_g g\cos\theta\,\partial_x h_l + \mathbf{S}_g \\ 0 \\ \mathbf{S}_d \end{bmatrix}.$$

Here A_k, Q_k, $u_k = Q_k/A_k$, and ρ_k are phase area, discharge, velocity, and density, with $A_g + A_l = A_f$. Subscripts g and l denote gas and liquid, and A_f is the undeformed full-pipe area.

Resolved gas is isothermal, $P_g = \rho_g RT = c_g^2\rho_g$, with $c_g = \sqrt{RT}$. The regular source vector $\mathbf{S}$ separates the geometric gas-pressure and hydrostatic coupling shown explicitly above from the axial-gravity, wall-friction, and interfacial-shear contributions collected in $\mathbf{S}_g$ and $\mathbf{S}_d$. The full constitutive correlations are deferred to the companion manuscript.

For a circular conduit of diameter D, the liquid central angle γ gives the liquid area $A_l = D^2(\gamma - \sin\gamma)/8$ and depth $h_l = D[1-\cos(\gamma/2)]/2$. The local free-surface width is $\partial A_l/\partial h_l = D\sin(\gamma/2)$, and the remaining gas area is $A_g = A_f - A_l$. Consequently, $\partial h_l/\partial A_l = (\partial A_l/\partial h_l)^{-1}$, which supplies the local hydrostatic geometry in both the stratified and vented free-surface limits.

The liquid restoring coefficient in Eq. (1) is

$$\Lambda_d = \frac{2gH_g}{A_l} + \frac{\rho_l - \rho_g}{\rho_l} g\zeta - \frac{\rho_g (u_g - u_l)^2}{\rho_l A_g}, \qquad \zeta = \frac{\cos\theta}{D\sin(\gamma/2)}. \tag{2}$$

H_g is gas-pressure head, θ is the pipe inclination, D is pipe diameter, and γ is the liquid central angle. The three terms represent gas pressure, hydrostatic–buoyancy restoration, and the inviscid Kelvin–Helmholtz slip contribution.

2.2. Multi-regime reductions of the decoupled system

The same area–discharge variables recover three hydraulic limits without changing the underlying definition of phase inventory. First, when $A_g = 0$ and $A_p \simeq A_f$, the liquid-full branch evolves $\mathbf{U}_p = [A_p, Q_p]^T$ according to

$$\begin{aligned} \partial_t A_p + \partial_x Q_p &= 0, \\ \partial_t Q_p + \partial_x \left(\frac{Q_p^2}{A_p} + gA_f h_s \right) &= \mathbf{S}_p, \end{aligned} \tag{3}$$

where $\mathbf{S}_p$ is the regular pressurized-branch momentum source and $\mathbf{R}_p = [0, \mathbf{S}_p]^T$ is the corresponding two-component source vector. Its pressure is supplied by the elastic-pipe closure

$$P_p - P_{\mathrm{ref}} = \rho_l a^2 \frac{A_p - A_f}{A_f}, \qquad h_s = \frac{P_p - P_{\mathrm{ref}}}{\rho_l g}, \tag{4}$$

where P_{ref} is the reference pressure, a is the water-hammer wave speed, h_s is the piezometric head, and $u_p = Q_p/A_p$ below denotes the liquid-full velocity.

Second, in a vented reach $P_g \simeq P_{\mathrm{atm}}$ and hence $H_g = \mathrm{const}$. With negligible gas inertia, the limiting relations $\rho_g/\rho_l \to 0$ and $\Lambda_d \to g\cos\theta/(\partial A_l/\partial h_l)$

hold, and the liquid equations take the following Saint–Venant form:

$$
\begin{aligned}
\partial_t A_l + \partial_x Q_l &= 0, \\
\partial_t Q_l + \partial_x \left(\frac{Q_l^2}{A_l} + \frac{g\cos\theta}{2(\partial A_l/\partial h_l)} A_l^2 \right) &= \mathbf{S}_d,
\end{aligned} \tag{5}
$$

The geometric derivative $\partial A_l/\partial h_l$ is frozen at the local state when the reduced flux is differentiated. Thus Eq. (5) is the locally frozen-coefficient Saint–Venant-type limit used by the companion algorithm. Third, on the stratified side of a nearly full ($A_l \to A_f$) moving front, the slip term is negligible but distributed gas mass and momentum remain governed by the first two components of Eq. (1). The liquid equations then take the near-front mixed-flow limit

$$
\begin{aligned}
\partial_t A_l + \partial_x Q_l &= 0, \\
\partial_t Q_l + \partial_x \left[\frac{Q_l^2}{A_l} + gA_f H_g + \frac{1}{2}\frac{\rho_l - \rho_g}{\rho_l} g\zeta A_l^2 \right] &= \mathbf{S}_d.
\end{aligned} \tag{6}
$$

This last relation is a local near-full reduction of the four-variable branch, not a separate two-equation replacement for the gas dynamics.

The stratified liquid subsystem is hyperbolic only for $\Lambda_d > 0$; the unmodified sign of Λ_d remains the instability diagnostic. For $\rho_g \ll \rho_l$ and negligible gas-pressure variation, the local inviscid critical-slip threshold is exceeded when

$$
(u_g - u_l)^2 > \frac{\rho_l - \rho_g}{\rho_g}\, g\zeta A_g. \tag{7}
$$

Beyond this threshold the inviscid short-wave problem is locally ill posed [12]. This sign change is an instability indicator, not a wave-breaking, entrainment, or liquid-bridge model; the face regularization introduced below does not alter that diagnostic. Thus the gas-acoustic, free-surface gravity-wave, elastic water-hammer, and moving mixed-flow-front responses belong to one parent architecture, although they do not share a single state dimension or pressure law. Lumped sealed-pocket and apparatus-specific pipe–shaft closures are declared separately; they are not additional parent-PDE regimes. The numerical subset and reductions used by each reported calculation are identified with that campaign.

2.3. Regime-transition Riemann problem

Let $x = x_\Gamma(t)$ denote a moving pressurized–stratified front, with $w_\Gamma = \dot{x}_\Gamma$. Pressurized flow is placed on the left and stratified flow on the right; reflection gives the reverse orientation. Only liquid mass and momentum enter the jump. The adjacent gas enters through its kinematic condition and the restoring coefficient $\Lambda_{d,\Gamma}$.

For a zero-thickness front with no liquid storage or concentrated liquid force, integration over a control volume that shrinks onto the front gives

$$\mathbf{F}_{p,\Gamma} - w_\Gamma \mathbf{U}_{p,\Gamma} - (\mathbf{F}_{f,\Gamma} - w_\Gamma \mathbf{U}_{f,\Gamma}) = \mathbf{0}, \tag{8}$$

where $\mathbf{U}_{p,\Gamma} = [A_{p,\Gamma}, Q_{p,\Gamma}]^{\mathrm{T}}$ and $\mathbf{U}_{f,\Gamma} = [A_{l,\Gamma}, Q_{f,\Gamma}]^{\mathrm{T}}$. Expanding Eq. (8) gives the liquid mass and momentum Rankine–Hugoniot conditions:

$$\begin{gathered} Q_{p,\Gamma} - Q_{f,\Gamma} = w_\Gamma \left(A_{p,\Gamma} - A_{l,\Gamma}\right), \\ \left(\frac{Q_{p,\Gamma}^2}{A_{p,\Gamma}} + g A_f h_{s,\Gamma}\right) - \left(\frac{Q_{f,\Gamma}^2}{A_{l,\Gamma}} + \frac{1}{2}\Lambda_{d,\Gamma} A_{l,\Gamma}^2\right) = w_\Gamma \left(Q_{p,\Gamma} - Q_{f,\Gamma}\right). \end{gathered} \tag{9}$$

The liquid-full branch cannot receive gas. The moving-frame gas condition is therefore

$$(Q_g - w_\Gamma A_g)_\Gamma = 0 \qquad \Longleftrightarrow \qquad u_{g,\Gamma} = w_\Gamma. \tag{10}$$

Here $A_{g,\Gamma} = A_f - A_{l,\Gamma}$. Bounded extrapolation from the nearest active stratified gas cells, together with the isothermal gas characteristic and state equation, provides $\rho_{g,\Gamma}$ and $P_{g,\Gamma}$; the nearest admissible state is used if extrapolation fails.

Because the water-hammer speed greatly exceeds the front speed, one pressurized acoustic characteristic reaches the front. Its compatibility relation is

$$u_{p,\Gamma} - u_{p,L} + \frac{g}{a}\left(h_{s,\Gamma} - h_{s,L}\right) = 0. \tag{11}$$

The reduced closure takes the stratified liquid trace from the adjacent state,

$$\left(A_{l,\Gamma}, Q_{f,\Gamma}\right) = \left(A_R, Q_R\right), \tag{12}$$

and freezes $\Lambda_{d,\Gamma}$ at that state.

Let $s_- = u_{l,R} - c_{l,R}$ and $s_+ = u_{l,R} + c_{l,R}$, where $c_{l,R} = [\max(\Lambda_{d,R}A_R, 0) + c_{\text{num}}^2]^{1/2}$. The admissible front-speed ranges are

$$\begin{aligned} \text{slow:} &\quad w_\Gamma \le s_-, \\ \text{middle:} &\quad s_- \le w_\Gamma \le s_+, \\ \text{fast:} &\quad w_\Gamma \ge s_+. \end{aligned} \tag{13}$$

After Eq. (12) fixes the stratified trace, the unknowns are w_Γ, $A_{p,\Gamma}$, and $Q_{p,\Gamma}$. Equations (9) and (11) determine them within each speed range. Positivity, speed ordering, and momentum-residual tolerance determine acceptance; bracketed iteration is retained as the fallback. The full residual construction and nonlinear solution are given in the companion methods paper.

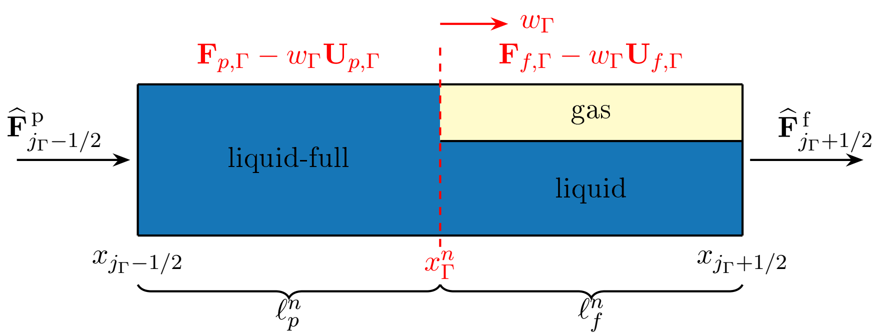

Figure 2: Control volumes and fluxes in a host cell cut by a moving pressurized–stratified front.

3. Numerical method

3.1. Finite-volume and front-tracking algorithm

Figure 2 shows the moving control-volume geometry used by the front-tracking update. Regular cells are updated in conservative finite-volume form,

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n - \frac{\Delta t}{\Delta x_j}\left(\widehat{\mathbf{F}}_{j+1/2} - \widehat{\mathbf{F}}_{j-1/2}\right) + \Delta t\,\mathbf{S}_j. \tag{14}$$

Fifth-order WENO-Z reconstruction is used in smooth regions [13]. Stencils adjoining a moving front, a boundary, or a newly formed regime use second-order minmod MUSCL–TVD reconstruction [14]; area and gas-density traces are limited toward their cell averages to preserve $0 < A_l < A_f$ and $\rho_g > 0$.

The two phases use branch-appropriate face problems rather than a single HLL or Rusanov flux. The gas flux consists of an area-aware quasi-one-dimensional isothermal-Euler entropy-conservative core and characteristic entropy dissipation [15, 16]. The liquid flux is sampled from a two-wave shallow-water Riemann solution [17], using a positive face-regularized restoring coefficient:

$$\widehat{\mathbf{F}}_g = \mathbf{F}_g^{\mathrm{EC}} - \tfrac{1}{2}\mathbf{K}_g(\mathbf{v}_{g,R} - \mathbf{v}_{g,L}),$$
$$\Lambda_d^h = \max\left[\tfrac{1}{2}(\Lambda_{d,L} + \Lambda_{d,R}), \frac{c_{\mathrm{num}}^2}{A_{l,\mathrm{ref}}}\right] > 0. \tag{15}$$

Here $\mathbf{F}_g^{\mathrm{EC}}$ is the entropy-conservative core, $\mathbf{v}_g$ is the gas entropy-variable vector, and $\mathbf{K}_g$ is a positive-semidefinite characteristic dissipation matrix. The regularization uses $A_{l,\mathrm{ref}} = \max[(A_{l,L} + A_{l,R})/2, A_\varepsilon]$, where $A_\varepsilon > 0$ is the area floor, and a positive local speed c_{num}. The area-aware gas construction prevents an area jump alone from diffusing gas mass. The positive Λ_d^h regularizes only the local face Riemann problem; the unmodified Λ_d remains the instability diagnostic. The liquid-full branch uses elastic-pipe speeds $u_p \pm a$. The path contribution associated with $P_g \partial_x A_g$ is balanced with the same reconstruction and quadrature.

Time integration uses second-order SSP–RK [18]. Reconstruction, branch fluxes, front traces, and sources are rebuilt at each stage. The step is limited simultaneously by liquid-full waves, stratified waves, the two cut subcells, and the time for the front to reach the next face, denoted by Δt_p, Δt_f, Δt_{cut}, and Δt_Γ, respectively:

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t\, \mathbf{L}(\mathbf{U}^n),$$
$$\mathbf{U}^{n+1} = \tfrac{1}{2}\mathbf{U}^n + \tfrac{1}{2}\left[\mathbf{U}^{(1)} + \Delta t\, \mathbf{L}(\mathbf{U}^{(1)})\right], \tag{16}$$
$$\Delta t = \min(\Delta t_p, \Delta t_f, \Delta t_{\mathrm{cut}}, \Delta t_\Gamma).$$

Here $\mathbf{L}$ is the semidiscrete spatial operator assembled from the reconstructed fluxes and sources.

When the front lies inside host cell j_Γ, the actual liquid-full and stratified lengths ℓ_p and ℓ_f are retained. For a control face moving at w_Γ, they satisfy

$\ell_p^{n+1} = \ell_p^n + w_\Gamma \Delta t$ and $\ell_f^{n+1} = \ell_f^n - w_\Gamma \Delta t$ when no face is crossed. With $\mathbf{U}_{p,\Gamma} = [A_{p,\Gamma}, Q_{p,\Gamma}]^T$ and $\mathbf{U}_{f,\Gamma} = [A_{l,\Gamma}, Q_{f,\Gamma}]^T$, define the relative liquid fluxes $\mathbf{G}_{k,\Gamma} = \mathbf{F}_{k,\Gamma} - w_\Gamma \mathbf{U}_{k,\Gamma}$, $k \in \{p, f\}$, and use the shared liquid flux $\widehat{\mathbf{G}}_\Gamma = \frac{1}{2}(\mathbf{G}_{p,\Gamma} + \mathbf{G}_{f,\Gamma})$. The subcells obey

$$\begin{aligned}
\ell_p^{n+1}\mathbf{U}_p^{n+1} &= \ell_p^n \mathbf{U}_p^n + \Delta t(\widehat{\mathbf{F}}_L^p - \widehat{\mathbf{G}}_\Gamma) + \Delta t\, \ell_p^n \mathbf{R}_p, \\
\ell_f^{n+1}\mathbf{U}^{n+1} &= \ell_f^n \mathbf{U}^n + \Delta t(\widehat{\mathbf{G}}_\Gamma^s - \widehat{\mathbf{F}}_R^f) + \Delta t\, \ell_f^n \mathbf{S}.
\end{aligned} \tag{17}$$

Here $\mathbf{U}$ and $\mathbf{S}$ in the second line are the complete four-component stratified state and source. In $\widehat{\mathbf{G}}_\Gamma^s$, the liquid entries equal $\widehat{\mathbf{G}}_\Gamma$, the gas-mass entry is zero, and the gas-momentum entry is $P_{g,\Gamma} A_{g,\Gamma}$, representing internal piston action. Adding the liquid parts cancels the shared relative flux, which gives geometric conservation as the front moves. The stratified update retains gas mass and momentum; gas disappears only through a recorded physical boundary flux.

If the front reaches a cell face, the stage is split at the exact arrival time and the remaining liquid and gas inventories are conservatively remapped to the new host cell. A remap producing nonpositive gas volume or density, or an excessive residual, is rejected and repeated with a smaller step; for a left-moving front, the two orientations are exchanged. Topology changes are tested only after a positive, conservative step has been accepted. A connected near-full run becomes a separate liquid-full segment only when

$$\min_{j_1 \le j \le j_2} \frac{A_{l,j}}{A_f} \ge \alpha_b, \qquad \sum_{j=j_1}^{j_2} \Delta x_j \ge L_{b,\min}. \tag{18}$$

Here $\alpha_b = 0.98$ and $L_{b,\min} = \max(D, 2\Delta x)$. This test identifies a candidate near-full run but does not prescribe its speed. The disclosed thresholds are numerical topology criteria, not a parameter-free wave-breaking or liquid-bridge model. Both bounding Regime-Transition Riemann Problems must be admissible, and segments merge only if the intervening gas has a connected escape path. Formation, merging, splitting, and release inherit the accepted liquid inventory and redistribute gas only within its connected region. A transaction that violates liquid mass, liquid momentum, gas inventory, or positivity is rejected. An under-resolved sealed pocket may use the lumped polytropic closure $P_g V_g^m = \text{const.}$; resolved stratified gas continues to use the distributed equations in Eq. (1), where m is the polytropic exponent.

3.2. Conservative pipe–shaft junction

The resolved pipe–shaft connection is treated as a massless, three-branch finite-volume node rather than as a prescribed source in one horizontal cell. The left, right, and upward branch coordinates point away from the junction, and their adjacent liquid traces are denoted by $(p_b^*, u_b^*, c_b, A_{l,b})$, where $b \in \mathrm{J} = \{L, R, \mathrm{UP}\}$. Here p_b^* and $u_b^* = Q_{l,b}^*/A_{l,b}$ are the pressure and outward-velocity traces obtained from the branch solution. The incoming liquid celerity is $c_b = a$ on an elastic liquid-full branch and $c_b = \sqrt{\Lambda_{d,b}^h A_{l,b}}$ on a stratified or free-surface branch, using the face-regularized coefficient defined in Eq. (15). The branch orientation and the two phase exchanges are shown in Figure 3.

For a trial absolute pressure p_J at the common node datum, each incoming

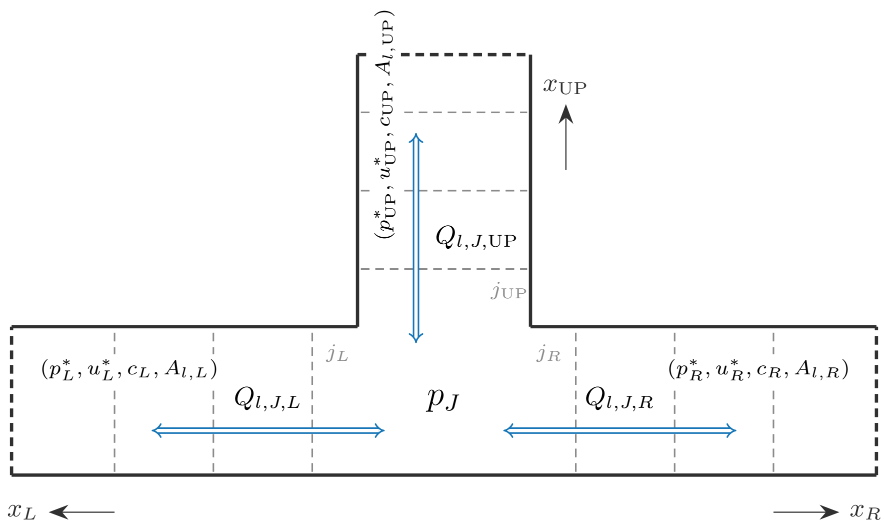

Figure 3: Pipe–shaft junction discretization. Grey dashed lines mark branch cell faces, p_J is a zero-storage node, and blue double-headed arrows denote signed liquid discharge.

characteristic supplies its own signed branch velocity and physical minor loss:

$$\begin{gathered}\frac{p_{l,J,b} - p_b^*}{\rho_l} = c_b\big(u_{J,b}^{\mathrm{out}} - u_b^*\big) + \frac{K_b}{2} u_{J,b}^{\mathrm{out}} \left|u_{J,b}^{\mathrm{out}}\right|, \\ p_{l,J,b} = p_J + \delta p_{h,b}, \qquad Q_{l,J,b}^{\mathrm{out}} = A_{l,b} u_{J,b}^{\mathrm{out}}, \\ \sum_{b \in \mathrm{J}} Q_{l,J,b}^{\mathrm{out}} = 0.\end{gathered} \tag{19}$$

Here $\delta p_{h,b}$ is the hydrostatic pressure offset between the common node datum and branch b, and K_b is its local loss coefficient. Because each characteristic relation is monotone in p_J, the scalar residual in the last line of Eq. (19) is bracketed and solved at every Runge–Kutta stage. No sign restriction is imposed on the resulting flows; water may therefore enter the shaft or return naturally to either horizontal branch. The three computed node fluxes replace the ordinary numerical fluxes at the corresponding branch faces. In the conservative pressure-potential gauge used by the liquid solver, the inserted face flux is

$$\widehat{\mathbf{F}}_{l,J,b} = \begin{bmatrix} Q^{\text{out}}_{l,J,b} \\ \left(Q^{\text{out}}_{l,J,b}\right)^2/A_{l,b} + \Psi^*_b + (p_{l,J,b} - p^*_b)A_{l,b}/\rho_l \end{bmatrix}. \tag{20}$$

Here Ψ^*_b is the adjacent branch value of the liquid pressure potential in the regular momentum flux $F_{Q_l} = Q_l^2/A_l + \Psi_l$. Thus $\Psi_{J,b} = \Psi^*_b + (p_{l,J,b} - p^*_b)A_{l,b}/\rho_l$ uses the same pressure gauge as the interior liquid operator. This pressure-difference form retains the full advective and pressure work but does not introduce an impulse into a uniform hydrostatic state.

Gas exchange is activated only when the resolved crown gas exposes a finite part of the shaft mouth. Its open area $A_{g,J}$ is obtained from the geometric overlap of the horizontal gas layer and the shaft bore. To distinguish the mouth problem from the area-weighted parent state in Eq. (1), let $\mathbf{q}^*_{g,b} = [\rho^*_{g,b}, \rho^*_{g,b}u^*_{g,n,b}]^\mathsf{T}$ denote the intrinsic gas state per unit mouth area, where $u^*_{g,n,b}$ is normal to the mouth. The per-unit-area isothermal gas Riemann flux $\widehat{\mathbf{f}}_g$ then supplies the normal mass and momentum exchange:

$$\begin{aligned} \begin{bmatrix} \dot{m}_{g,J} \\ \dot{I}_{g,\text{UP},J} \end{bmatrix} &= A_{g,J}\widehat{\mathbf{f}}_g\left(\mathbf{q}^*_{g,h}, \mathbf{q}^*_{g,\text{UP}}\right), \\ \Delta M_{g,h,J} &= -\Delta t\, \dot{m}_{g,J}, \\ \Delta M_{g,\text{UP},1} &= +\Delta t\, \dot{m}_{g,J}, \\ \Delta J_{g,\text{UP},1} &= +\Delta t\, \dot{I}_{g,\text{UP},J}. \end{aligned} \tag{21}$$

Here $M_{g,h,J}$ is the cell-integrated gas mass in the horizontal junction cell, whereas $M_{g,\text{UP},1}$ is that in the first vertical cell. The quantity $J_{g,\text{UP},1}$ is the vertical gas-momentum inventory. Accordingly, $\dot{m}_{g,J}$ is the gas mass rate from the horizontal branch into the shaft and $\dot{I}_{g,\mathrm{UP},J}$ is its resolved vertical momentum-flux rate. The horizontal axial velocity is tangential to this mouth, so its normal gas trace is zero; the vertical trace comes from the first shaft cell. Gas mass is subtracted from the horizontal inventory and added to the shaft inventory in one residual evaluation. Momentum remains directional at the 90° turn: $\dot{I}_{g,\mathrm{UP},J}$ updates vertical momentum, horizontal axial momentum retains its own advective flux, and their vector difference is the reaction on the fitting. It would therefore be incorrect to transfer one scalar momentum flux unchanged between the two axes. Donor positivity is enforced by time step subdivision rather than by prescribing a transfer fraction. Both phase exchanges are internal conservative fluxes, not post-step source terms.

Equations (19)–(21) define the reference conservative junction closure used by the model family. Finite-geometric-node, fitted-shaft-core, and finite-mouth slug extensions are identified with the corresponding experiment; a reduced archive is not described retroactively as having executed every component of this junction solver or of the numerical method above.

4. Experimental assessment

The assessment comprises three campaigns based on published geysering experiments. Simulations start from the documented experimental initial conditions, except for the explicitly identified C9 phase-one all-water variant used in the analytic comparison. Shaft motion, gas release, and regime selection are outcomes of the production calculations. Outside the explicitly labelled Test A sensitivity and the disclosed Test B screen, the calculations use no prescribed release rate, scripted geyser height, overflow accumulator, or per-case observation fit.

The three campaigns provide complementary evidence. Campaign 1 (Section 4.1) reproduces the water-filled ventilation-shaft experiments of Vasconcelos and Wright [2], a widely used controlled benchmark for geysering models. It tests time-resolved histories in both regimes: Test A represents the non-geysering branch, and Test B represents the geysering branch. Campaign 2 (Section 4.2) reproduces the horizontal-pipe/vertical-riser experiments of Cong, Chan and Lee [3]. Because that study condenses its outcomes into an explicit two-parameter geysering criterion, this campaign tests the model as a regime classifier across a systematic parameter sweep rather than against individual histories. Campaign 3 (Section 4.3) addresses the prototype-motivated junction-chamber experiments of Liu, Shao and Zhu [5]. These experiments connect the laboratory benchmarks to a field-scale drop-shaft geometry. Other studies are used as corroborating references rather than reproduction targets. They include the field observations of [1], the larger-scale releases designed for three-dimensional CFD calibration [4], the covered-manhole geysers of [6], the mitigation study of [7], and the vertical-pipe violent geysers of [19]. Their geometries are either insufficiently controlled for a one-dimensional comparison, or their focus (cover dynamics, mitigation devices, or flashing-driven eruption) lies outside the entrapped-air release mechanism considered here.

4.1. Campaign 1: water-filled ventilation shafts (Vasconcelos and Wright, 2011)

Vasconcelos and Wright [2] used a horizontal acrylic pipe of diameter $D = 0.094$ m with both far ends closed. A ventilation tower of length $L = 0.610$ m and variable diameter D_t was mounted on a coupling 3.516 m from the upstream end. A butterfly valve separated a 0.546-m-long upstream air chamber from the water-filled reach (Fig. 1). The air chamber was pressurized to an initial gauge head H_{a0}, and the tower was partially filled to an initial level $Y_{fs,0}$. Each run began when the valve was opened by hand in less than one second. The pressurized pocket then propagated along the pipe crown toward the tower and vented through the water column. A pressure transducer was located 1.07 m downstream of the valve. The experimental programme varied $D_t \in \{57.1, 44.4, 25.4, 12.7\}$ mm ($D_t/D = 0.607$–0.135), $H_{a0} \in \{0.305, 0.610, 0.915\}$ m, and $Y_{fs,0} \in \{0.254, 0.356, 0.457\}$ m. Tower diameter controlled the observed response: for $D_t/D = 0.607$, the free-surface rise never exceeded 10% of L, whereas for $D_t/D = 0.135$, the free surface reached the tower top in every repetition.

Two configurations are selected to span the two regimes (Table 1). Following [2], results are non-dimensionalized by the tower length and the tower gravity-velocity scale: elevations and pressure heads as $Y^* = Y/L$ and $H^* = H/L$, and time as $T^*_{\mathrm{rel}} = t\sqrt{gD_t}/L$. Geysering is defined as the free surface Y^*_{fs} reaching the tower top ($Y^* = 1$) before the rising air–water interface Y^*_{int} catches the free surface.

The one-dimensional model geometry follows the apparatus exactly: upstream air chamber 0.546 m (the initial pocket volume, 3.79×10^{-3} m^3), mid-

Table 1: Selected test configurations from the experimental matrix of [2].

Test	D_t (mm)	D_t/D	H_{a0} (m)	$Y_{fs,0}$ (m)	$L/\sqrt{gD_t}$ (s)	Observed regime
A	57.1	0.607	0.305	0.356	0.815	no geysering
B	12.7	0.135	0.610	0.356	1.728	geysering (3/3)

dle pipe 2.970 m, downstream closed pipe 0.490 m, tower at $x = 3.516$ m, and the transducer sampled at $x = 1.616$ m. The initial pocket pressure is $P_{a0} = P_{\text{atm}} + \rho_l g H_{a0}$; the pipe downstream of the valve and the tower below $Y_{fs,0}$ are initially at rest and water-filled. The experimental curves used for comparison were digitized from the published figures of [2] (pressure head from Figs. 5 and 6; free-surface and interface trajectories from Figs. 7 and 8, three experimental repetitions each). Pressure repetitions are summarized by a representative pointwise curve, while level repetitions are redrawn individually as vector markers.

4.1.1. Test A: wide tower, non-geysering branch ($D_t/D = 0.607$)

Test A is the wide-tower, non-geysering member of Campaign 1. It first tests whether the reduced model selects venting without overflow. It then compares the computed pressure and tower-level scales with the measurements. In the experiment, the trapped pocket vents through the tower before the free surface reaches the rim. During interface rise, Y^*_{fs} remains close to 0.6, and the transducer head retains a long plateau until pocket exhaustion [2]. The archived one-dimensional and planar OpenFOAM calculations select the same branch. Two complementary archives support the comparison. Figure 4 examines common-clock event ordering and phase topology. Figure 5 quantifies a separately declared connected-pocket closure sensitivity. The two figures are not presented as outputs from one numerical realization.

The nearest-time states in Fig. 4 divide the event into four stages. Near $t = 1.50$ s, both calculations show the early post-valve transient while the pocket remains in the horizontal pipe. Near $t = 6.90$ s, gas has reached the junction and entered the water-filled tower. Near $t = 7.60$ s, the breakthrough transition is accompanied by rapid drainage and phase redistribution in the tower. The one-dimensional liquid-equivalent height has fallen to 0.150 m, while the planar field resolves the irregular junction-scale breakup that a cross-section-averaged column cannot represent. By $t = 10.10$ s, the one-dimensional archive retains only 0.0155 m of liquid-equivalent height, concentrated visibly below approximately 0.082 m above the tower base; the planar field retains a broader, laterally deformed residual-liquid region. The paired 1D and 2D archives are sampled without a time shift, with a maximum time mismatch of 0.020 s. They preserve horizontal-pocket propagation, tower entry, and breakthrough ordering and remain on the non-geysering branch, but the late phase distribution is not matched. The snapshots therefore support branch selection and event ordering, not a quantitative phase-field error norm. For legibility only, 1D tower cells with $\alpha_l \leq 0.02$ are omitted from the raster rendering; their liquid volume remains included in the reported equivalent height.

The native-time histories in Fig. 5 provide the quantitative comparison. The one-dimensional history is an explicitly isolated sensitivity. It combines the contact-preserving release update with a connected-pocket pressure-relaxation closure of time scale $\tau_b = 0.10$ s. This optional closure probes unre-

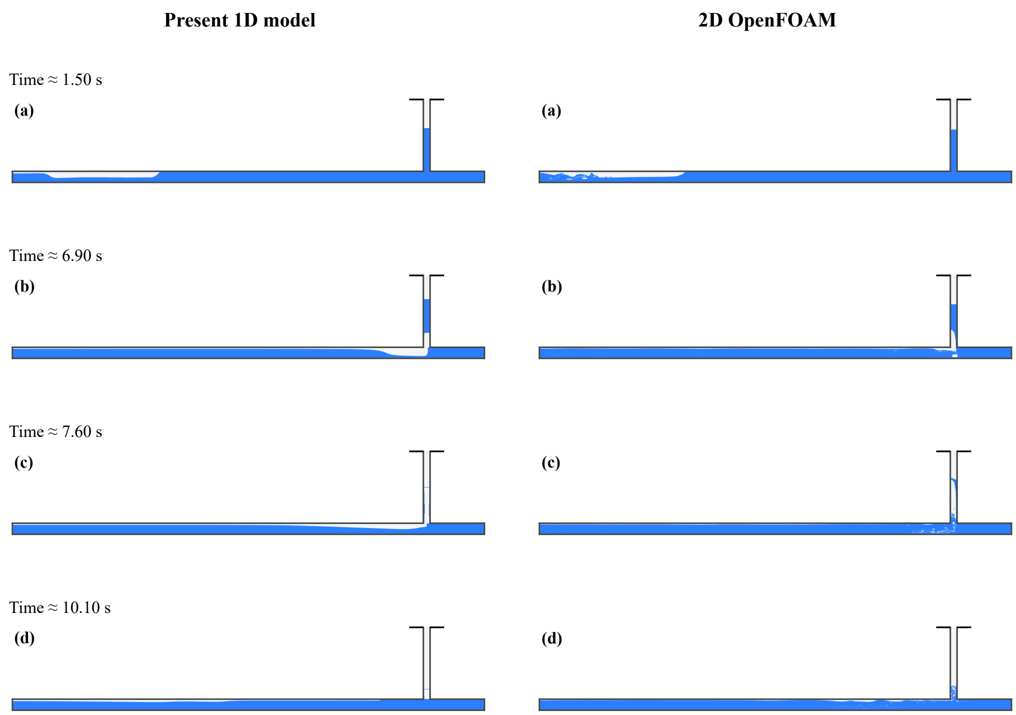

Figure 4: Test A phase evolution near $t = 1.50$, 6.90, 7.60, and 10.10 s: present 1D model (left) and planar OpenFOAM simulation (right). The corresponding archived 1D times are 1.480, 6.908, 7.608, and 10.108 s, respectively; no time shift is applied. Blue denotes water and white denotes gas. Row (c) shows the breakthrough transition and rapid tower drainage, and row (d) shows the small residual-liquid region near the tower base. In the 1D rendering only, cells with $\alpha_l \leq 0.02$ are visually omitted while their liquid volume remains in the equivalent-height audit.

solved exchange between the coherent pocket and the draining wall film, and it is not used for the other campaign claims. The values below characterize the disclosed sensitivity rather than an independently validated production closure.

The pre-venting pressure scale is close to the experiment, but the terminal release remains late. Over $4 \leq T^*_{\mathrm{rel}} \leq 7.5$, the one-dimensional plateau median is $H^* = 0.539$, compared with 0.537 for the extracted experimental central trace. After hydrostatic conversion from the probe elevation, the crown-referenced two-dimensional mean is 0.562 over $1 \leq T^*_{\mathrm{rel}} \leq 7$. This datum conversion is not a fitted offset. The main discrepancy lies in the release clock. The experimental, two-dimensional, and one-dimensional traces first fall below $H^* = 0.30$ at $T^*_{\mathrm{rel}} = 8.28$, 8.63, and 9.20, respectively. The one-dimensional sensitivity therefore preserves the plateau but delays terminal depressurization by about 0.92 dimensionless time units. No time or pressure shift is applied.

The short one-dimensional spike near $T^*_{\mathrm{rel}} = 9.18$ coincides with the archived level-coalescence event. It is retained for provenance but is not interpreted as a resolved physical pressure rebound or included in the plateau statistic.

All three records remain below the rim, and the one-dimensional level-event times fall within the experimental spread. The one-dimensional free surface rises gradually to $Y^*_{fs,\max} = 0.613$. The largest digitized experimental marker is 0.649, and the two-dimensional maximum is 0.623; none reaches the rim. The one-dimensional interface crosses $Y^*_{int} = 0.05$ at $T^*_{\mathrm{rel}} = 7.85$. The raw archive reaches exact equality with the free surface at 9.18, whereas the displayed trace stops at the one-cell criterion at 9.13. Across the three digitized repetitions, the first markers above the same $Y^*_{int} = 0.05$ threshold span approximately 7.36–7.94. Piecewise-linear crossings of the digitized interface and free-surface tracks span approximately 8.85–9.19. Both one-dimensional event times lie within the observed run-to-run spread, although the sensitivity does not reproduce the fastest interface rise. The sensitivity captures the non-geysering outcome, pressure plateau, and maximum free-surface scale, but terminal depressurization remains late. The remaining shape differences are consistent with unresolved coherent-core/wall-film exchange. These data do not identify that mechanism uniquely.

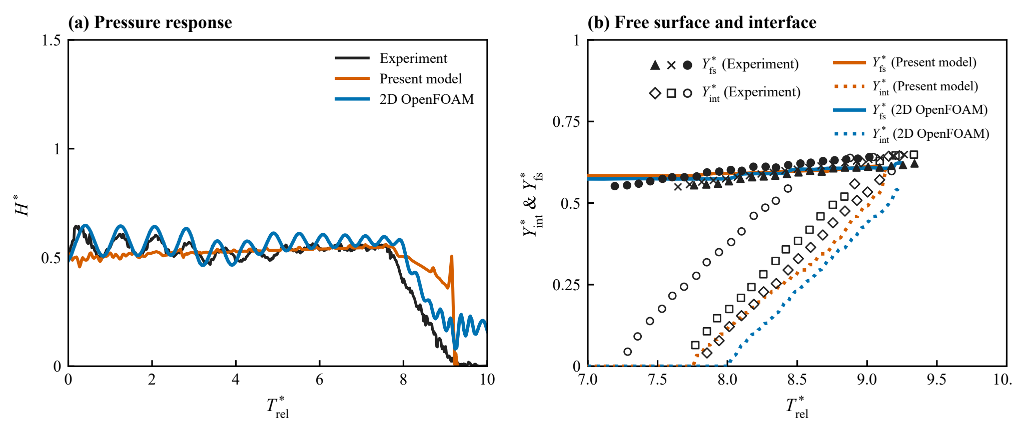

Figure 5: Test A comparison with [2]: (a) pressure head and (b) normalized free-surface and interface elevations. Black denotes the experiment, orange the present 1D model, and blue OpenFOAM.

The planar OpenFOAM calculation is a supporting spatial diagnostic rather than a geometry-exact reference. In the vertical plane, the tower-to-pipe area ratio is $D_t/D = 0.607$. The circular apparatus instead has $A_t/A_p = (D_t/D)^2 = 0.369$. The planar model also cannot reproduce a three-dimensional annular wall film. Quantitative assessment is therefore anchored to the experiment. Within that scope, Test A supports the wide-tower branch and the bulk pressure and water-level scales, while revealing a systematic late bias in terminal pocket depressurization. The interface event times remain within the digitized three-run spread. This non-geysering reference provides the controlled contrast for the narrow-tower Test B; it is not an independent validation of the optional connected-pocket closure.

4.1.2. Test B: small tower, geysering branch ($D_t/D = 0.135$)

The narrow tower provides the geysering counterpart to Test A. Vasconcelos and Wright reported a geyser in every repetition. In the three Fig. 8 records, gas first appears in the tower over $T^*_{\mathrm{rel}} = 3.648$–$3.742$. The free surface reaches the rim over $T^*_{\mathrm{rel}} = 3.900$–$3.981$ and therefore precedes gas breakthrough. The credible high-interface markers occur over $T^*_{\mathrm{rel}} = 4.09$–$4.17$. The Fig. 6 transducer record has a pre-release plateau near $H^* = 0.76$ and a collapse envelope of approximately $T^*_{\mathrm{rel}} = 4.02$–$4.18$ [2]. The $D_t/D = 0.135$ velocities in the source Table 2 are averages over the tower-diameter group rather than measurements for this centre-panel condition, so they are used only as contextual scales. The source Fig. 11 is likewise not imported: although it uses the same tower diameter, its initial head and water level are 0.305 and 0.254 m, respectively, and therefore define a different test.

The quantitative one-dimensional history uses a crown-datum pressure and a pressure-driven tracked-front application closure. Its Test B mouth resistance and coverage parameters were selected from a predeclared six-point bounded screen: $K_{\mathrm{slug}}/K_{\mathrm{g}} = 1.25$, 1.50, or 1.75, and $\alpha^{\mathrm{full}}_{g,\mathrm{j}} = 0.10$ or 0.12. The admissible 1.50/0.10 pair minimized the native interface error. The resulting errors are in-sample closure diagnostics rather than an independent calibration test. No time shift, curve fit, or level-trajectory smoothing is introduced. The spatial snapshots serve a different role from the quantitative history. Figure 6 uses a separate visualization archive. Its horizontal state is Tosan-based, and its tower is reconstructed from saved Y_{fs}, Y_{int}, and jet-height scalars. It supports morphological comparison rather than quantitative tower-field validation. The planar OpenFOAM calculation also serves only as an independent spatial diagnostic. It retains the published pipe lengths and initial heads, but uses the area-equivalent slit width $W = D_t^2/D$ and neglects surface tension. It is a controlled two-dimensional surrogate for the release pathway, not a geometry-exact representation of the 12.7-mm circular tower.

The quantitative one-dimensional archive selects the observed geysering branch. The standard two-dimensional archive selects the same branch. In the one-dimensional result, the shock-fitted gas front crosses $Y_{int}^* = 0.05$ at $T_{\mathrm{rel}}^* = 3.622$. The conservative free-surface observer reaches the predeclared one-cell rim-completion level at $T_{\mathrm{rel}}^* = 3.923$, where $1 - \Delta z/L = 0.984$. This occurs before the computed front/free-surface catch at $T_{\mathrm{rel}}^* = 4.051$. The corresponding two-dimensional entry and rim events occur at $T_{\mathrm{rel}}^* = 3.85$ and 4.11. Both calculations therefore place rim arrival ahead of the resolved upper-interface trajectory, although the two-dimensional surrogate develops the sequence later. The archived two-dimensional window does not resolve a subsequent gas-core catch, and none is inferred from this comparison. The three rows in Fig. 6 compare the complete pipe–tower domain at the same physical times without event alignment: gas-pocket approach at $t = 6.35$ s, tower entry and water-column lift at 7.25 s, and the initial discharge/venting response at 7.70 s. This common clock exposes the timing difference; the rows do not supply the event metrics quoted above.

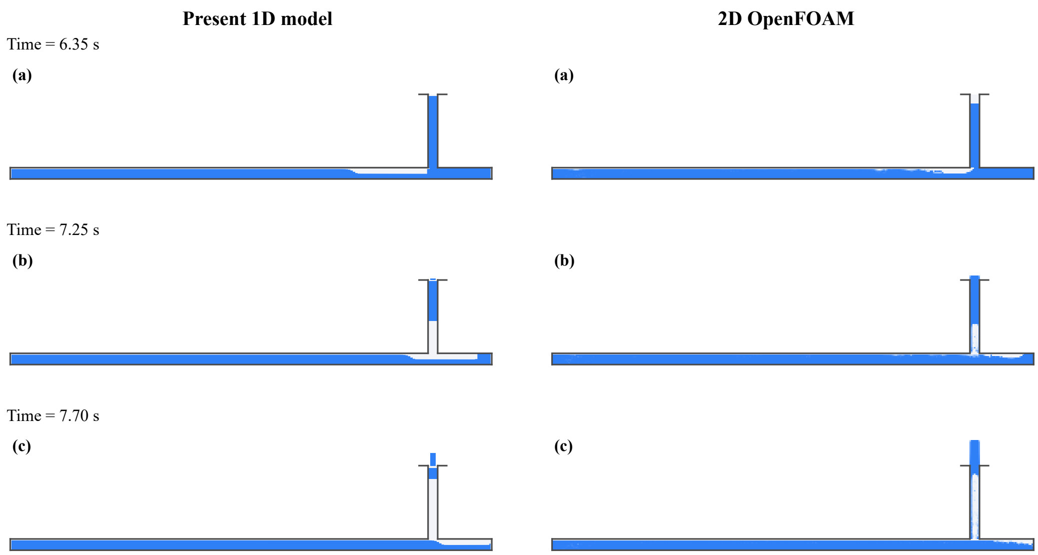

Figure 6: Test B phase evolution at t = 6.35, 7.25, and 7.70 s: present 1D model (left) and planar OpenFOAM simulation (right). Blue denotes water and white denotes gas.

The native-time histories in Fig. 7 separate close one-dimensional agreement in the bulk scales from larger event-timing and two-dimensional biases. The experimental pre-eruption stand is $Y^*_{fs} = 0.823$, compared with 0.834 in the one-dimensional model and 0.869 in the two-dimensional surrogate. At the crown datum, the one-dimensional pressure plateau is $H^* = 0.754$, versus the digitized experimental pseudo-median value 0.758 used in Table 2. The black curve itself is the pointwise arithmetic mean of the three digitized repetitions. The standard raw two-dimensional pre-release mean is higher, $H^* = 0.838$ over $1 \leq T^*_{\mathrm{rel}} \leq 3$. The displayed 0.40-s one-dimensional pressure mean falls below $H^* = 0.30$ at $T^*_{\mathrm{rel}} = 4.097$, within the experimental collapse envelope. The standard two-dimensional head crosses the same threshold at 4.823. All of these comparisons retain the native time coordinate. For the shape-only comparison in Fig. 7(a), the blue pressure history comes from a separate full-open-rim two-dimensional archive. It is displayed after one constant offset, $\Delta H^* = -0.104$; no time shift or shape fit is applied. This curve is excluded from the raw quantitative metrics. All quoted two-dimensional pressure values and Table 2 use the standard, unshifted-amplitude archive.

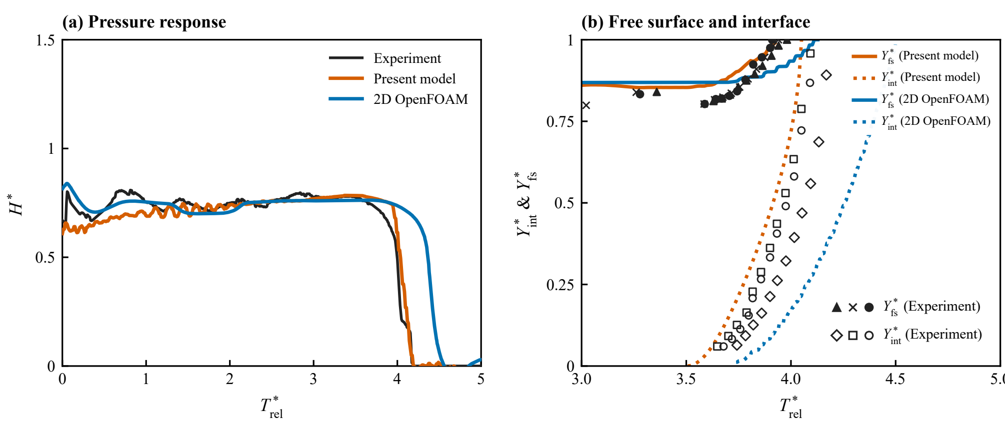

Figure 7: Test B comparison with [2]: (a) normalized pressure head and (b) normalized free-surface and interface elevations. Black denotes the experiment, orange the present 1D model, and blue OpenFOAM.

The level histories recover the rising curvature, but the tracked interface remains too early and high. The isolated digitized point at $(T^*_{\mathrm{rel}}, Y^*) = (3.847, 0.92)$ is retained in the plot for provenance. It is not used as an arrival estimate because it lies outside all three rising-interface tracks. Zero-level baseline/sentinel symbols outside the rising tracks are also excluded from the trajectory comparison and are not treated as positive-height measurements. With these symbols excluded, the native-time rising-track RMSEs are 0.042 and 0.206 for the one-dimensional Y^*_{fs} and Y^*_{int} histories, respectively; their signed biases are +0.034 and +0.172. Both computed tracks accelerate over the rising bands, with late-to-early slope ratios of 1.37 and 1.96. The corresponding two-dimensional RMSEs remain 0.052 and 0.303. The revised native curvature therefore removes the former piecewise-linear shape error, but not the early, high interface bias. The connected cellwise $\alpha_g = 0.50$ field front reaches only $0.016L$ in this calculation. Accordingly, Y^*_{int} denotes the computed shock-fitted front closure rather than a resolved dense-core isocontour. No plotting or digitization offset is used to remove the remaining separation.

Table 2 reports only quantities that can be compared without differentiating sparse video markers. The scalar experimental pressure values are read from the digitized pseudo-median; the reported event intervals retain the visible three-run scatter. Both calculations reproduce the outcome and event ordering. Relative to the experimental intervals, the one-dimensional tracked front begins slightly early and remains too high, but the grid-complete rim event and pressure collapse fall within the measured spans. The two-dimensional surrogate enters slightly late, spills late, and retains its excess head longest.

Table 2: Native-time, unshifted-amplitude Test B validation summary. Experimental intervals are digitized three-run spans from the centre panels of Figs. 6 and 8 in [2]; scalar experimental values use the digitized pseudo-median, whereas the black pressure curve in Fig. 7(a) is the pointwise three-run mean. The 2D column uses the standard raw archive and excludes the display-offset blue pressure curve. The one-dimensional pressure-collapse row uses the same 0.40-s centered mean shown in panel (a); its level-event rows use native samples.

Observable	Experiment	1D	2D
Outcome	geyser, 3/3	geyser	geyser
Pre-release H^*	0.758	0.754	0.838
Pre-entry Y^*_{fs}	0.823	0.834	0.869
$Y^*_{int} = 0.05$ crossing, T^*_{rel}	3.648–3.742	3.622	3.848
First rim arrival, T^*_{rel}	3.900–3.981	3.923[a]	4.114
Pressure pseudo-median $H^* < 0.30$, T^*_{rel}	4.048	4.097	4.823

[a]The one-dimensional rim event uses the predeclared one-cell completion criterion $Y^*_{fs} \geq 1 - \Delta z/L = 0.9836$; the topology latch and plotted geometric observer cross it at the same native sample.

Across the two tests, the archived branch evidence selects venting without overflow in the wide tower and eruption in the narrow tower. The Test A connected-pocket sensitivity recovers the pressure and free-surface scales but delays terminal depressurization; its interface event times remain within the three-run spread. It is not conflated with the separate spatial reconstruction. For Test B, the quantitative one-dimensional history closely reproduces the pre-eruption pressure and water-column scales, while the shock-fitted interface remains early and high. The grid-complete rim event and pressure collapse lie within the experimental spans. The standard planar calculation supports the same entry-to-overflow morphology, but overpredicts the pre-release head and water level and delays the principal events. Quantitative comparison remains anchored to the experiment. The two-dimensional results and display reconstructions provide supporting evidence subject to the limitations stated in their captions. Section 5 relates the remaining timing pattern to the shaft-mouth and entrainment closures.

4.2. Campaign 2: entrapped-air release from a horizontal pipe into a vertical riser (Cong, Chan and Lee, 2017)

The apparatus of [3] consists of a horizontal acrylic pipe with diameter $D = 0.05$ m and length approximately 6 m. Its upstream end is connected to a constant-head tank, and a tee-junction supports a vertical riser of height 1.8 m and variable diameter ($D_r = 16$–46 mm). An air pocket of prescribed initial length L_0 is created in the horizontal pipe between ball valves. Release under the upstream head H_0 drives the pocket along the pipe crown to the riser, where it vents through the initially water-filled column. Video and high-speed cameras recorded the air–water interface trajectories in the pipe and riser. Transducers measured pressure near the pipe end and at the riser base (Fig. 8).

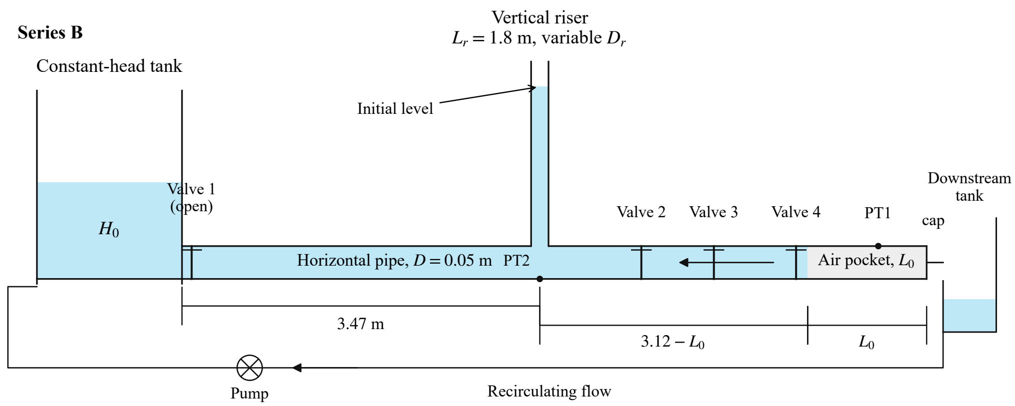

Figure 8: Campaign 2 Series-B apparatus, redrawn from Fig. 1(b) of [3].

The authors summarized the outcomes of their parameter study with a two-parameter geysering criterion,

$$\frac{D_r}{D} \leq 0.62 \qquad \text{and} \qquad V^*_{air} = \frac{V_{air}}{(\pi D_r^2/4)\, H_0} \geq 3.42, \tag{22}$$

which requires both a sufficiently narrow riser and a pocket volume large relative to the riser water column.

This campaign evaluates the model as a regime classifier over a systematic sweep, rather than against individual time histories as in Campaign 1. The first comparison treats the seven high-speed-camera runs of the experimental Series B individually (B-H1 – B-H7: $H_0 = 0.66$ m, $L_0 = 0.61$ m, riser diameters $D_r/D = 0.32$–0.92). Computed classification, pocket arrival time, and mean rise velocity are compared with the measurements. The second comparison covers a broader synthetic sweep over the experimental parameter ranges ($D_r = 16$–46 mm, $L_0 = 0.61$–1.8 m, $H_0 = 0.66$–0.88 m; 63 configurations). The resulting classifications are compared with the experimental criterion in Eq. (22).

All campaign runs use the same fully synchronous junction coupling as Campaign 1 and the identical frozen solver configuration (the Campaign 1 Test B closure set). No parameter is changed between campaigns or between runs. Two geometric features are specific to this apparatus. The upstream boundary is the constant-head tank, and the pocket is created *downstream* of the riser tee. The released pocket therefore travels upstream along the crown toward the riser. The tee position, 2.88 m from the tank, is fixed from the experiment's kinematics. The measured arrival times and front speeds in Table 2 of [3] give a valve-to-tee distance $T_a U_f$ of 2.51/1.92/1.32 m for the three pocket lengths $L_0 = 0.61/1.2/1.8$ m, all placing the tee at the same station. This position is held fixed across the sweep. An earlier campaign version used a staged horizontal/vertical treatment whose classifier reduced to a static volume–diameter criterion. Those results were circular as a validation of Eq. (22) and are superseded by the fully coupled results reported here.

4.2.1. Series B: classification, arrival times, and rise velocities

The model reproduces five of the seven measured Series-B classifications (Table 3 and Fig. 9). It predicts geysering for B-H1 ($D_r/D = 0.32$) and no geysering for B-H4–B-H7 ($D_r/D = 0.62$–0.92), with no false positives. The two intermediate geysering runs, B-H2 and B-H3 ($D_r/D = 0.42$, 0.52), are misclassified as non-geysering. Their computed free surfaces rise to 0.87 and 0.82 m but do not reach the rim. The error is one-sided: the model under-predicts eruption. Section 5 examines how volume-neutral mouth exchange may contribute to this bias. The same exchange allows the narrow-riser pocket to rebuild the overpressure that powers the B-H1 geyser, but it also cancels part of the one-to-one column lift in the reservoir-fed layout as riser width increases. The computed arrival times are late by 1.5 s on average, with the largest delay in the widest riser where the arrested crown current covers the final reach slowly. The measured spread is 7.84–8.22 s.

The rise-speed comparison reveals an overly pulsed computed response. In the geysering run B-H1, the computed eruption-window climb speeds (v_{fs} = 1.71, v_{int} = 1.28 m/s as the fastest sustained 0.6-s climb) exceed the respective measured whole-event means (0.92, 1.23 m/s). The interface speed is close, but the free-surface speed is markedly high. The eruption jet is therefore over-sharp, consistent with the narrow-riser bias found in Test B of Campaign 1. In the no-geysering branch, the computed gas-nose climb speeds (0.69–1.20 m/s) exceed the measured 0.42–0.48 m/s. The computed climb is concentrated in surge pulses rather than the observed steady Taylor-scale ascent. The computed free-surface response (0.20–0.65 m/s during the pulses, zero sustained rise) has the same pulsed character, compared with the measured steady 0.2–0.26 m/s.

The synchronous coupling recovers the measured ranking of the pipe-end pressure transients. The geysering run B-H1 develops a cycle-averaged compression surge to 1.71 H_0, compared with the reported value of about 1.9 H_0. The no-geysering run B-H6 shows only a gentle post-arrival hump to 1.02 H_0, compared with the reported value of about 1.4 H_0. The staged treatment gave both runs the same 2.07 H_0 peak. The synchronous coupling instead recovers the experimental differentiation: a wide riser vents the pocket and thereby damps the pipe-end pressure. This gain is accompanied by the two intermediate-diameter classification misses discussed above.

Table 3: Campaign 2, experimental Series B of [3] ($H_0 = 0.66$ m, $L_0 = 0.61$ m): measured against computed classification, pocket arrival time T_a, maximum computed free-surface height $Y_{fs,\max}$ (riser top at 1.8 m), and rise speeds of the free surface (v_{fs}) and gas nose (v_{int}). Measured speeds are whole-event means (Table 2 of [3]); computed speeds are the fastest sustained 0.6-s climb in the venting window. Misclassified runs are marked †.

			Geysering		T_a (s)			v_{fs} (m/s)		v_{int} (m/s)
Run	D_r/D	V^*_{air}	Meas.	Model	Meas.	Model	$Y_{fs,\max}$ (m)	Meas.	Model	Meas. / Model
B-H1	0.32	9.03	yes	yes	8.07	7.56	1.80	0.92	1.71	1.23 / 1.28
B-H2†	0.42	5.24	yes	no	7.84	–	0.87	0.77	–	1.02 / –
B-H3†	0.52	3.42	yes	no	8.18	9.56	0.82	0.66	0.43	0.92 / 1.24
B-H4	0.62	2.40	no	no	8.14	9.40	0.82	0.21	0.53	0.42 / 1.20
B-H5	0.72	1.78	no	no	8.10	9.88	0.86	0.26	0.65	0.48 / 1.05
B-H6	0.82	1.37	no	no	8.10	9.16	0.78	0.25	0.39	0.48 / 1.05
B-H7	0.92	1.09	no	no	8.22	11.44	0.79	0.20	0.20	0.44 / 0.69

The two signature runs clarify how riser diameter changes the pressure and water-column response (Fig. 10). These runs are also examined as individual histories in Sections 4.2.1 and the case folders. Their archived reproductions include per-run comparisons with the digitized level and pressure histories from Figs. 7, 9 and 10 of [3]. In B-H1 ($D_r/D = 0.32$), the released pocket arrives at 7.56 s (measured 8.07 s). The entering gas drives the free surface upward in two surge pulses, while arrest of the supply-line counter-flow compresses the pocket to 1.71 H_0 (cycle-averaged; reported $\approx 1.9\,H_0$). The resulting column ejection places the free surface at the rim while the gas nose remains below it, which is the geysering signature. In B-H6 ($D_r/D = 0.82$), the same release delivers the same pocket, but the wide

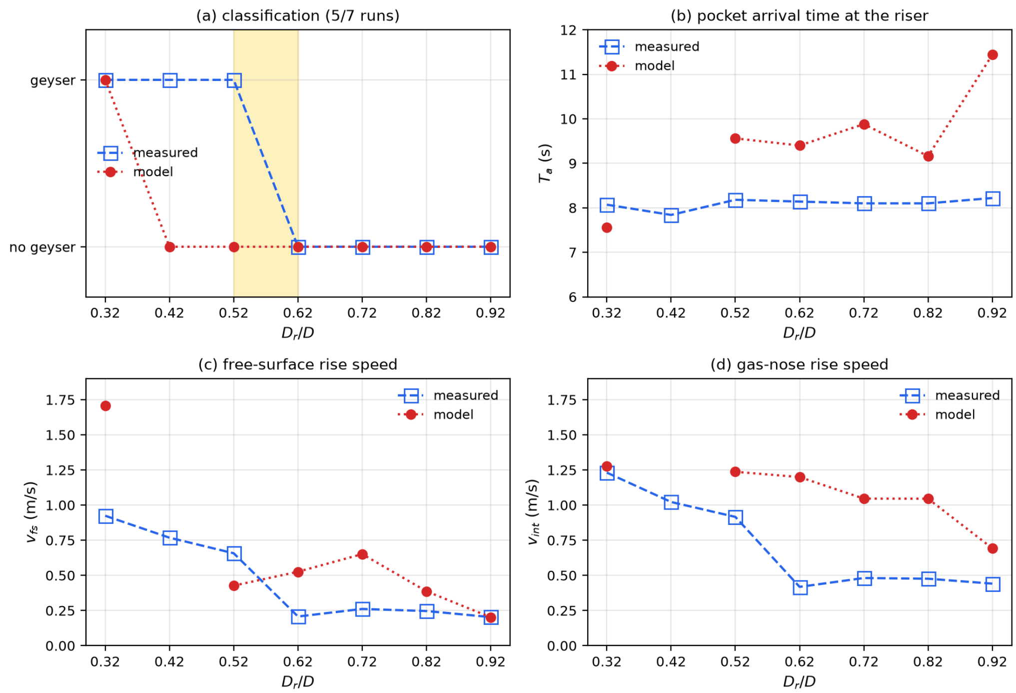

Figure 9: Campaign 2 Series B [3]: (a) geysering classification, (b) pocket-arrival time, (c) free-surface rise speed, and (d) gas-nose rise speed versus D_r/D.

riser passes the gas with only a gentle free-surface rise to 0.78 m. The pocket vents without eruption, and its pressure never exceeds 1.02 H_0 after arrival. Because only D_r changes, the pipe-end pressure provides the experiment's cleanest controlled contrast. The model reproduces its sign and ranking, although both pressure levels are under-predicted. The 1.4 H_0 no-geyser hump is missed because the computed free-surface lift is under-predicted in the reservoir-fed layout (Section 5).

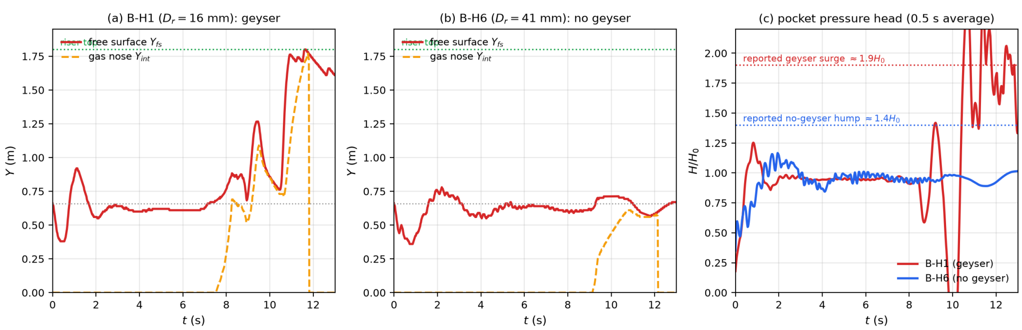

Figure 10: Campaign 2 signature cases [3]: free-surface and gas-nose trajectories for (a) B-H1 and (b) B-H6, and (c) their pocket-pressure heads.

The common-clock spatial comparison for B-H1 supports the same branch but exposes different eruption clocks (Fig. 11). At the physical times 8.5, 12.8, and 13.0 s, the frozen 1D archive reports a much earlier free-surface rise. It reaches 1.79 m above the pipe crown by 13.0 s, whereas the refined planar OpenFOAM field still places the water column below the 1.8 m rim. The 2D calculation reaches 98% of the rim at 13.865 s and resolves water above 2.1 m at 14.8 s. Both descriptions select the measured geysering branch, but their eruption clocks and spatial representations differ. The 2D gas-arrival time, 8.50 s, is close to the measured 8.07 s. Its subsequent eruption is nevertheless late relative to the reported 9.55 s, and its post-arrival pressure surge is not reproduced. The sequence therefore provides supporting evidence for branch selection and morphology, not a quantitative OpenFOAM validation.

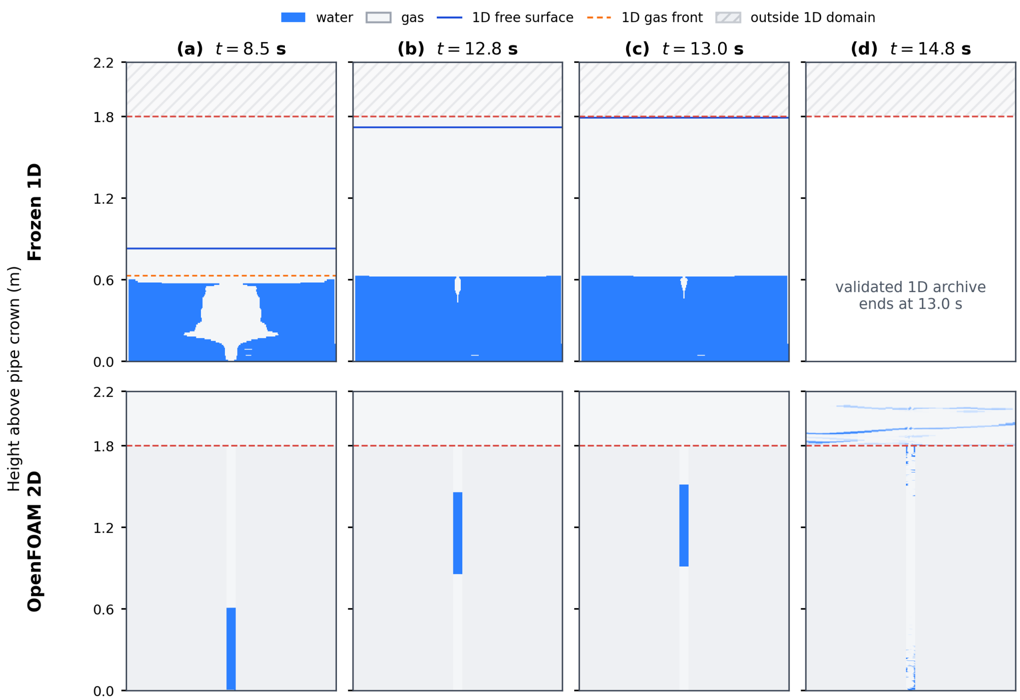

Figure 11: B-H1 riser phase sequence at $t = 8.5$, 12.8, 13.0, and 14.8 s: present 1D model (top) and planar OpenFOAM simulation (bottom). The archived 1D run ends at 13.0 s.

The complete-domain comparison for B-H6 preserves the observed non-geysering sequence at the photographic clock (Fig. 12). The three times correspond to gas arrival (8.7 s), developed rise (9.9 s), and late venting (10.9 s). Each row places the published B-H6 riser photograph between the full 1D and planar OpenFOAM domains, with no event alignment. The 1D reconstruction passes a compact gas pocket with weaker water-column lift, whereas the 2D VOF field gives a stronger and more spatially distributed late response. Because the high-speed record images the riser rather than the full horizontal pipe, the photographic column validates the riser sequence. The outer panels show the corresponding system-level phase distribution.

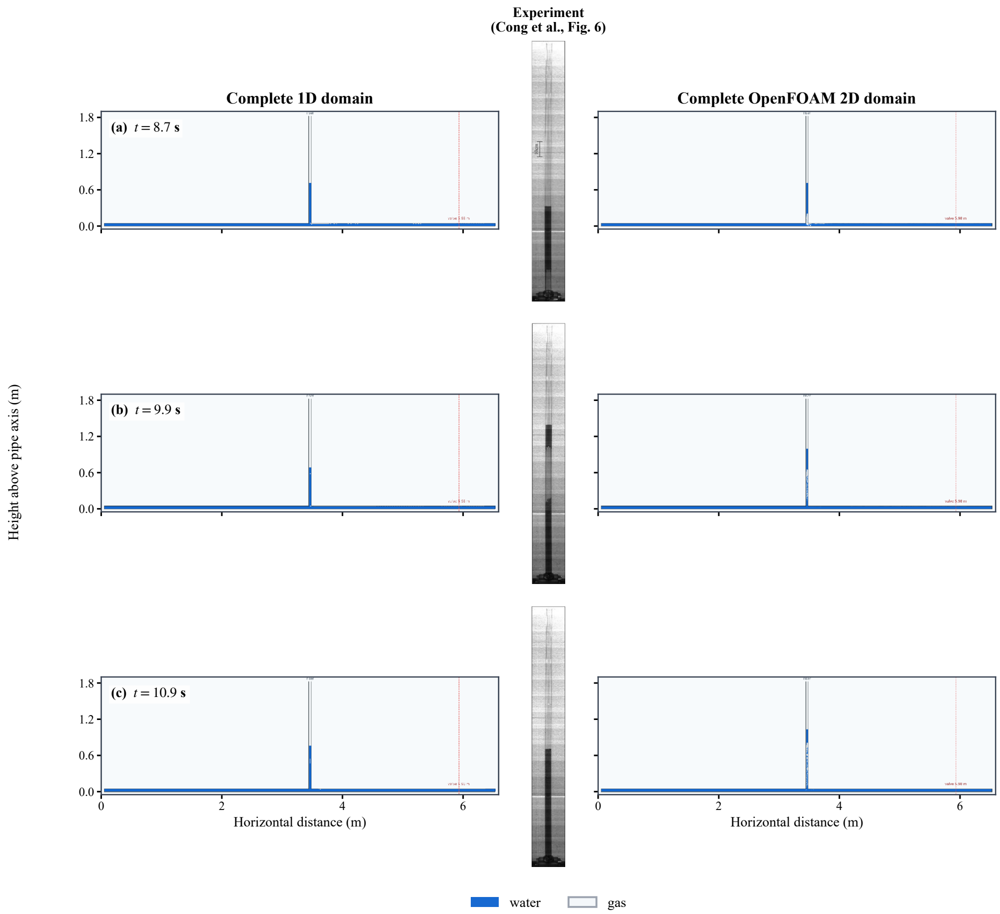

Figure 12: B-H6 full-domain comparison at (a) 8.7, (b) 9.9, and (c) 10.9 s: present 1D model (left), experiment [3] (centre), and planar OpenFOAM simulation (right).

The aggregate Series-B comparison conceals how the two descriptions differ within an individual non-geysering event. Figure 13 therefore overlays the digitized B-H6 markers from Fig. 7(a) of [3] with a geometry-matched one-dimensional application and the area-equivalent planar OpenFOAM calculation. All traces retain physical time; only the dense two-dimensional trace receives a centred 0.05-s running median for display. Both calculations reproduce the observed non-geysering outcome. The two-dimensional gas reaches the riser at 8.04 s, close to the measured 8.10 s, whereas the one-dimensional arrival is 8.66 s. The trajectory comparison is less complete: free-surface marker RMSE is 0.097 m for the two-dimensional result and 0.310 m for the one-dimensional result, while gas-nose RMSE is 0.245 and 0.317 m, respectively. Both under-predict the final lift, but the spatial calculation substantially reduces the free-surface error.

The focused one-dimensional trace is a geometry- and valve-matched application variant whose horizontal gas front uses the Benjamin-celerity closure; it is shown to diagnose this case and does not replace the frozen Series-B values in Table 3. Likewise, the planar calculation is supporting evidence for spatial consistency rather than an expansion of the primary validation scope.

A source-matched comparison preserves the experimental parameter context by transcribing the B-H6 row of Table 2 in [3] (Table 4). The second block reports only quantities available robustly from all three records. This makes the outcome and arrival-time comparison explicit without substituting peak-based simulation values for the source table's whole-event velocity definitions.

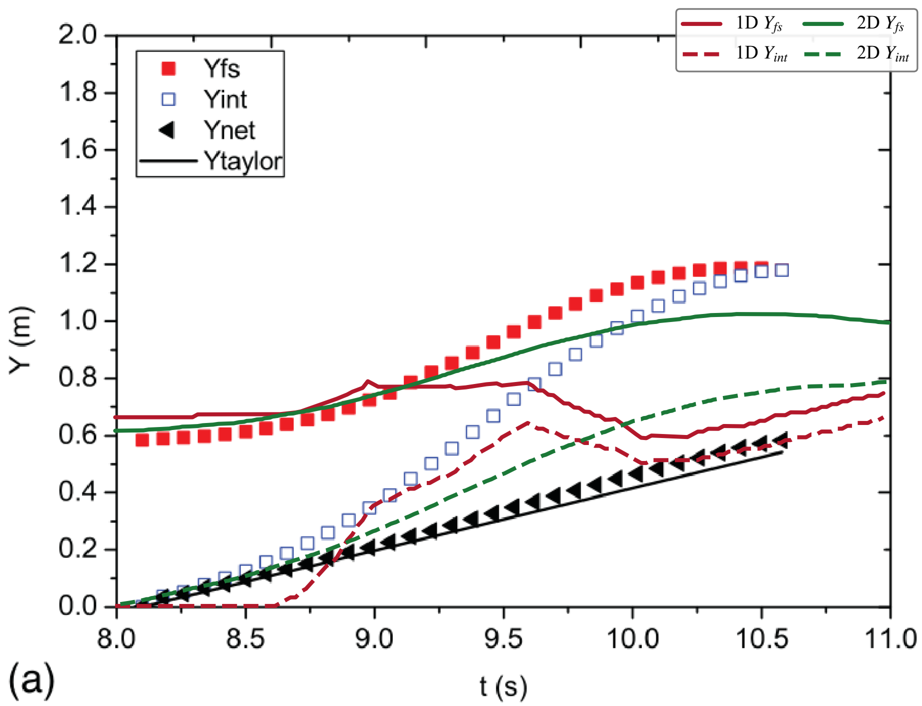

Figure 13: B-H6 free-surface (Y_{fs}, solid) and interface (Y_{int}, dashed) trajectories overlaid on Fig. 7(a) of [3]. Dark red and green denote the present 1D model and OpenFOAM, respectively.

Table 4: Source-matched B-H6 parameter and result comparison. The first block transcribes the B-H6 row of Table 2 in [3]; the second block adds the experimental, 1D, and OpenFOAM 2D outcome, gas-arrival time, peak levels, and marker RMSE. Dashes denote quantities not defined for the experimental reference.

Published B-H6 condition and whole-event measures			
Parameter	Value	Parameter	Value
D_r (m)	0.041	H_0 (m)	0.66
L_0 (m)	0.61	T_a (s)	8.10
$U_f/\sqrt{gD}$	0.443	D_r/D	0.82
V_{air}/V_w	1.37	Geyser	No
$\overline{v}_{fs}$ (m/s)	0.246	$\overline{v}_{int}$ (m/s)	0.476
$\overline{v}_{net}$ (m/s)	0.235	v_{Taylor} (m/s)	0.219

Source	Outcome	T_a (s)	$Y_{fs,\max}$ (m)	$Y_{int,\max}$ (m)	RMSE Y_{fs} (m)	RMSE Y_{int} (m)
B-H6 experiment	No	8.10	1.201	1.178	–	–
1D	No	8.66	0.786	0.700	0.310	0.317
OpenFOAM 2D	No	8.04	1.021	0.940	0.097	0.245

4.2.2. Regime map against the experimental criterion

The 63-configuration sweep yields a one-sided missed-eruption bias in the criterion plane (D_r/D, V^*_{air}) (Fig. 14). Each point represents one full synchronous simulation classified without knowledge of Eq. (22); the criterion lines are drawn only for comparison. The model agrees with the experimental criterion in 39 of 63 configurations. In all 24 disagreements, the criterion predicts geysering and the model does not. There are no false positives. Every modelled eruption lies inside the criterion's geysering region. The no-geysering half-plane $D_r/D > 0.62$ and the small-volume region $V^*_{air} < 3.42$ are classified correctly throughout. The misses concentrate at intermediate diameters ($D_r/D = 0.42$–0.62) combined with large pocket volumes ($L_0 \geq 1.2$ m). In these cases, the computed free surface rises to 1.0–1.6 m but remains below the 1.8-m rim; seven of the 24 reach within 0.45 m of it. This pattern extends the Series-B under-lift bias across the parameter plane. The model retains both end members and the interaction that eruption requires narrow risers *and* large relative pocket volume, but requires more restrictive conditions for eruption and under-predicts it in the transition band. For screening, the bias direction is consequential: the frozen model misses geysers near the boundary rather than producing false alarms.

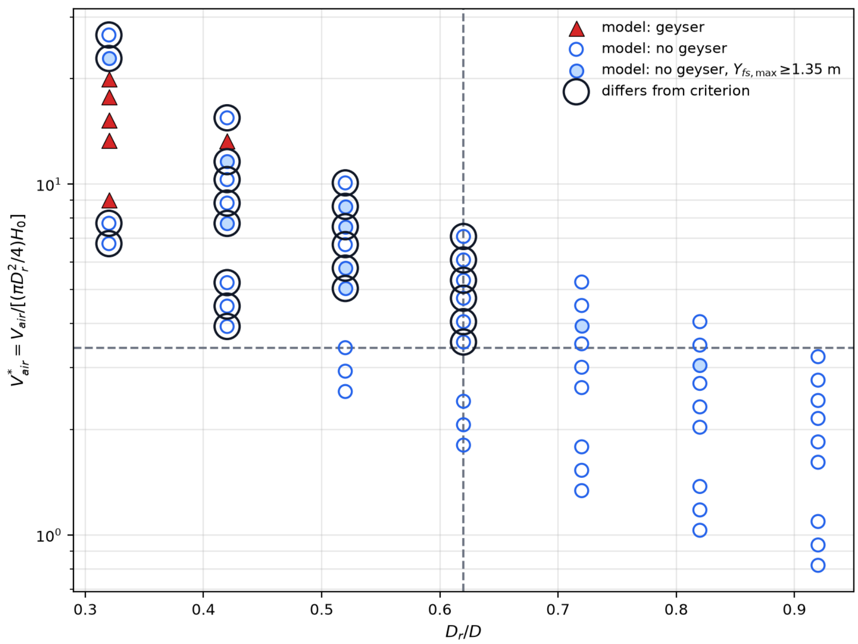

Figure 14: Campaign 2 classification of 63 configurations in the criterion plane of Eq. (22). Filled triangles denote model geysering, circles non-geysering, and black rings disagreement with the experimental criterion.

4.3. Campaign 3: stormwater geyser in a vertical shaft above a junction chamber (Liu, Shao and Zhu, 2020)

The apparatus of [5] is a 1:20-scale model of a 27 m deep Edmonton prototype dropshaft. A 5.80 m upstream pipe ($D = 0.20$ m, slope 1:100) feeds a transparent junction chamber ($0.30 \times 0.30 \times 0.45$ m) with an invert drop of 0.18 m to a 5.95 m downstream pipe ($D_d = 0.28$ m). A vertical shaft of diameter $d_r = 0.06$ m and height 1.22 m is mounted at the chamber lid centre and vents to the atmosphere. Inflow is imposed as a flow step $Q_0 \to Q_1$ with valve opening time $T_v \approx 0.4$ s. Pressure transducers PT1–PT4 monitor the shaft, the chamber lid (PT2, the primary datum), the chamber floor, and the upstream crown (Fig. 15).

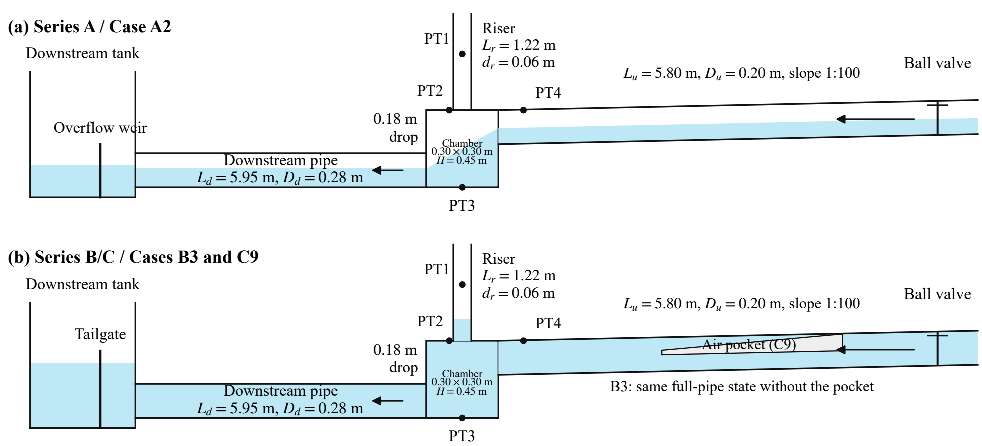

Figure 15: Campaign 3 apparatus and initial states, redrawn from Fig. 2 of [5].

The three configurations use the documented baseline numerical settings from the primary Campaigns 1–2 quantitative archives. The horizontal reach is a single composite one-dimensional domain comprising the upstream pipe, chamber, and downstream pipe. Internal momentum faces represent the chamber connections, and a rigid-column ordinary differential equation for the shaft water column is coupled at the lid tap. This reduced formulation does not resolve three-dimensional chamber mixing, slug venting at the chamber gas port, or the late-time arrival of a compressing air pocket along the upstream crown at the level of the companion two-fluid/RTRP framework. These limitations are stated explicitly where they affect the comparison.

4.3.1. Branch pair A2/B3: downstream boundary flips the outcome

Cases A2 and B3 share the same inflow step ($Q_0 = 20$, $Q_1 = 100$ L/s) and differ only in the downstream tail: A2 discharges to an open channel (weir-controlled, $h_d/D_d = 1/4$), whereas B3 is initially surcharged with a submerged tail gate. In the experiment A2 never geysered while B3 produced a single-shoot eruption on every repetition (Fig. 5 of [5]).

The unchanged model reproduces the A2/B3 branch flip. For A2, the riser column remains below the rim ($h_{r,\max} = 0.47$ m against a shaft height of 1.22 m). The steady chamber pressures agree with the digitized Fig. 3 record: PT3 is 5.26 kPa against 4.99 kPa (the model is about 5% high), and PT2 is 1.80 kPa against 2.15 kPa (the model is about 16% low in the standing-column head that emerges from the composite domain rather than being prescribed). The computed B3 run geysers, with the free surface reaching the shaft top and overflowing. Its eruption timing is close, with peak PT2 at 1.68 s against 1.47 s. However, the frozen elastic branch celerity $a_{wh} = 40$ m/s underpredicts the compressive peak amplitude: the PT2 peak is 31 kPa against 55 kPa (−44%). A sensitivity run at $a_{wh} = 80$ m/s gives 63.5 kPa, showing that the peak is highly sensitive to acoustic wave speed. This single sensitivity test does not exclude a contribution from closure error. At the baseline computed peak pressure, the experimental regression in Fig. 7(a) gives approximately 2.50 m ($h = 0.694\, P_{\max}/\rho g + 0.309$ m, $R^2 = 0.97$). The computed jet height is 2.17 m, about 0.33 m lower.

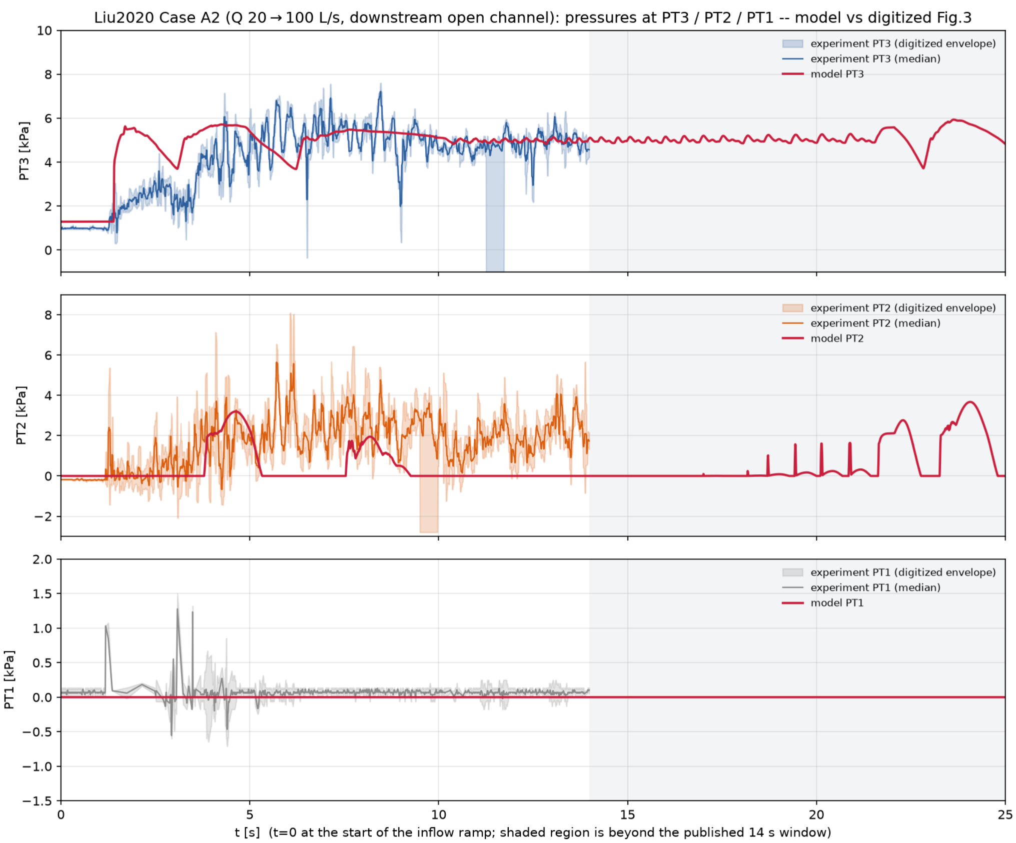

Figure 16: Case A2 (Q: 20→100 L/s, downstream open channel): chamber and shaft pressures against time. Blue: digitized experimental PT2/PT3 from Fig. 3 of [5]; red: present model.

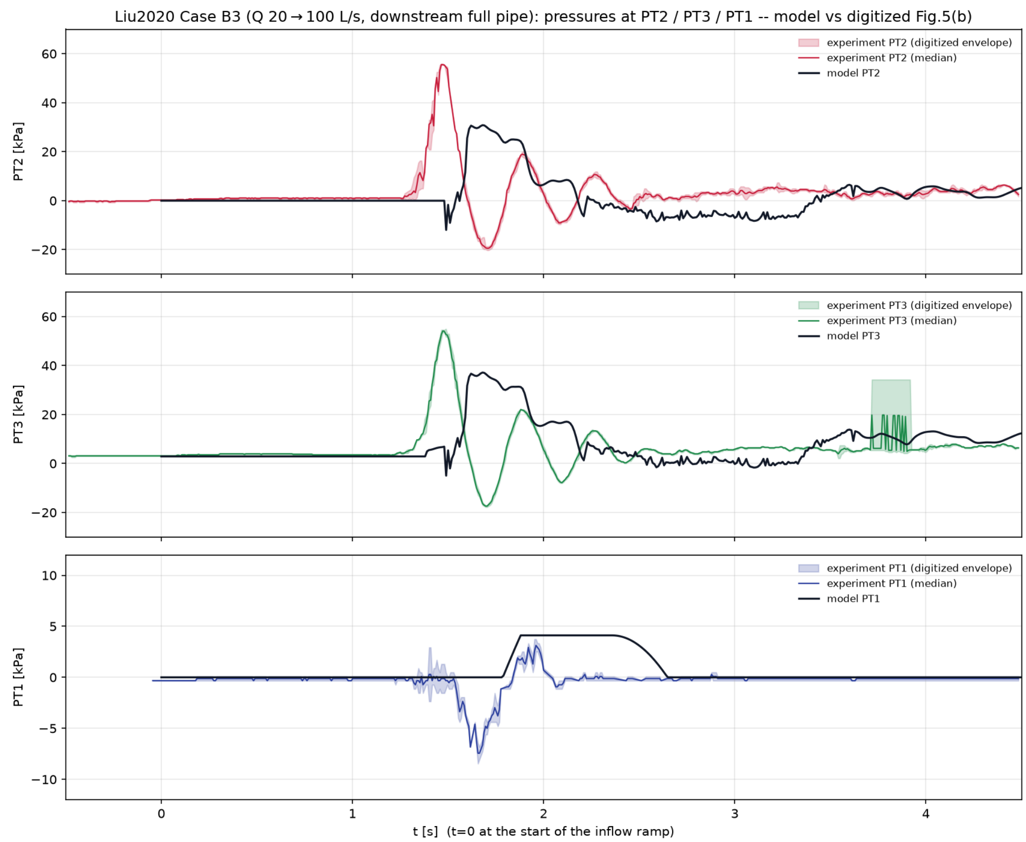

Figure 17: Case B3 (Q: 20→100 L/s, downstream surcharged): PT1–PT3 pressure histories. Blue: experiment [5]; red: present model.

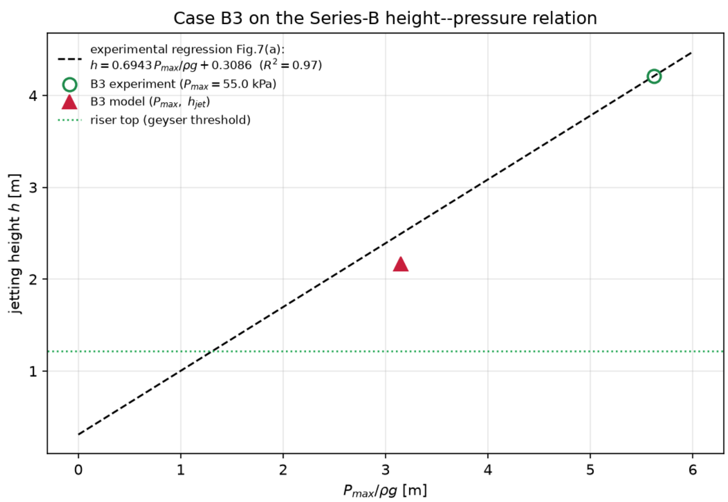

Figure 18: Case B3 on the eruption-height–peak-pressure diagram of Fig. 7(a) of [5]. Open symbols: experimental family; filled red circle: present model ($h = 2.17$ m, $P_{\max} = 31$ kPa at $a_{wh} = 40$ m/s).

4.3.2. Case C9: transient-driven geysers and the analytic mass-oscillation model

Case C9 ($Q_0 = 25 \rightarrow Q_1 = 40$ L/s, surcharged downstream tail, initial shaft column $h_{r0} = 0.30$ m, sealed air wedge on the upstream crown in the experiment) is the authors' detailed two-stage case: the first two geysers are driven by the hydraulic transient before the trapped pocket reaches the chamber at $t \approx 6.46$ s; geysers 3–8 are pocket-driven and lie outside the present reduced solver.

For the phase-one comparison the pocket is omitted in the computation (the all-water variant assumed in Eq. (7) of [5]), while the sealed-pocket run is retained as a qualitative cushioning check. The analytic head at the chamber lid is

$$H_s(t) = h_{r0} + \frac{L_d}{gA_d}\frac{\mathrm{d}Q_u}{\mathrm{d}t}\left(1 - \cos\frac{2\pi t}{T}\right), \qquad T = 2\pi\sqrt{\frac{L_d A_r}{gA_d} + \frac{h_{r0}}{g}}, \quad (23)$$

with $T = 1.52$ s for C9.

The all-water model captures the first C9 pressure peak in both timing and magnitude (Fig. 19). The figure compares the digitized PT2 record from Fig. 9, converted to head at the chamber-top datum, with Eq. (23) and the all-water model over the first 5 s. The measured first peak is $P_{1m} = 10.69$ kPa at $t = 0.50$ s, and the model gives 11.5 kPa at 0.54 s. The free surface reaches the shaft top at 0.73 s in the experiment and 0.93 s in the model. The oscillation period is $T = 1.45$ s in the paper (Eq. (8)), compared with 1.13 s in the model and 1.52 s in Eq. (23). The model period is closer to the analytic value than to the measured period, but remains fast, consistent with the stiff one-dimensional tail-gate coupling. After the first transient, the steady heads agree within $\approx 20\%$: the experimental PT2 final value is 8.79 kPa, and the model gives 10.6 kPa. The sealed-pocket variant qualitatively delays and cushions the first peak as in the experiment. It then drifts after ≈ 4 s as the wedge compresses through the reduced gas model, a phase-two process outside the stated scope.

Across the three cases, the composite-domain formulation extends the frozen Campaign 1–2 solver to a prototype-motivated junction geometry. The A2/B3 pair reproduces the change in regime caused by the downstream boundary, and C9 phase 1 reproduces the transient mass-oscillation geyser. Agreement covers regime selection and timing, as well as amplitude where the governing response is hydraulic rather than acoustic. The main residuals are the under-resolution of short compressive pressure spikes with the frozen

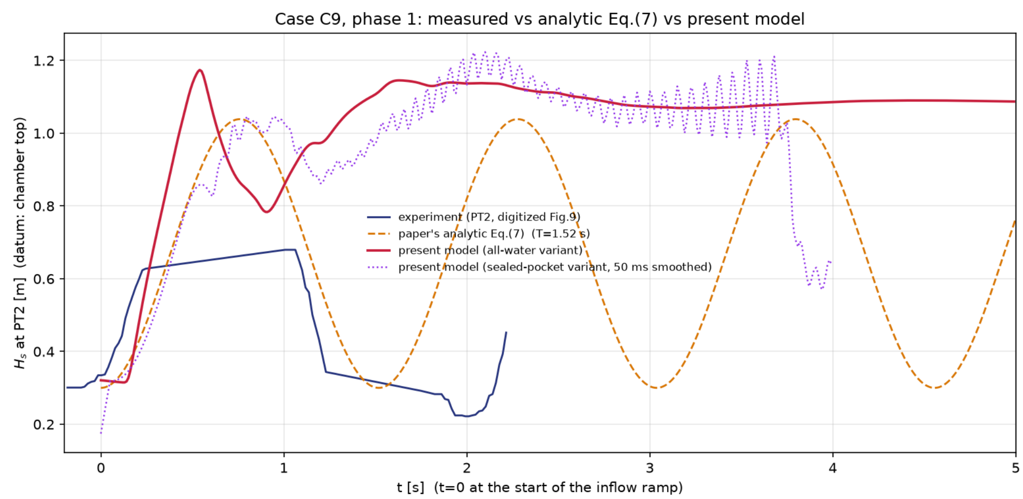

Figure 19: Case C9, phase 1 ($t < 5$ s): PT2 head H_s. Blue: experiment [5]; orange dashed: Eq. (23); red: all-water model; purple dotted: sealed-pocket variant.

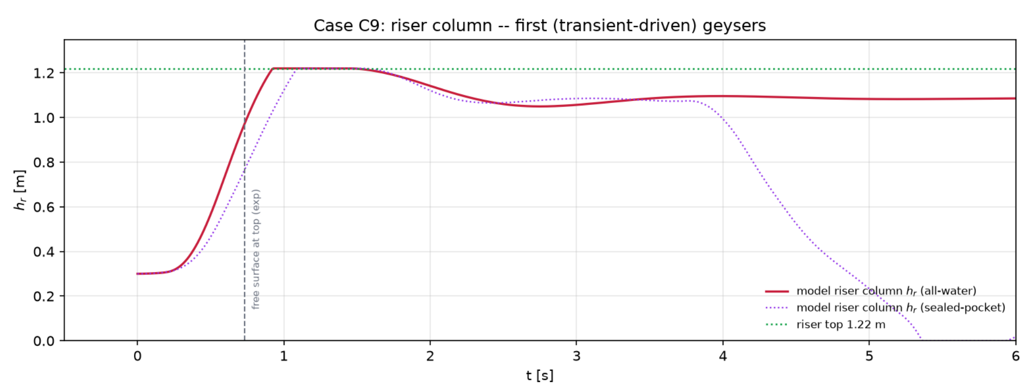

Figure 20: Case C9: shaft water-column height h_r for the all-water (solid) and sealed-pocket (dotted) variants. The dashed line marks the experimental time at which the free surface first reaches the shaft top in phase 1.

Table 5: Campaign 3 summary metrics. Acoustic peaks use the frozen $a_{wh} = 40$ m/s; no coefficient is retuned between cases.

Quantity	A2 (no geyser)	B3 (geyser)	C9 phase 1
Geyser?	no / no	yes / yes	yes / yes
PT2 steady or peak [kPa]	1.80 / 2.15	31 / 55	11.5 / 10.7
Peak or top-arrival time [s]	–	1.68 / 1.47	0.54 / 0.50
Shaft top reached [s]	–	≈ 1.65	0.93 / 0.73
Oscillation period [s]	0.30 (PT2–PT3)	–	1.13 / 1.45

Slash-separated cells: model / experiment; single values are identified by row.

elastic celerity and the deliberate omission of pocket-transport physics after the pocket reaches the chamber.

5. Discussion

5.1. Comparison across the three experimental data sets

Each data set stresses a different part of the model. Campaign 1 examines event evolution, Campaign 2 examines flow-regime classification, and Campaign 3 examines the response to the downstream chamber boundary. Campaigns 1–2 show that explicit gas conservation and transport can recover the observed pattern of venting versus geysering. Campaign 3 instead tests the reduced hydraulic branches against a boundary-driven regime change and an initial all-water transient. Matching these outcomes does not guarantee accurate event timing, interface velocity, or short-duration pressure peaks.

In Campaign 1, the model selects the observed non-geysering outcome for the wide tower and the geysering outcome for the narrow tower. The wide-tower evidence comes from two separate calculations. The visualization-only reconstruction preserves the non-overflow event order, but its tower empties faster than the planar calculation in the last common frame. The independently identified connected-pocket sensitivity gives a pressure plateau of $H^* = 0.539$, while the extracted experimental central trace gives 0.537. Its maximum free-surface elevation is $Y^*_{\mathrm{fs,max}} = 0.613$; the largest experimental marker is 0.649, and the planar value is 0.623. The interface lift-off and free-surface catch times are 7.85 and 9.18, respectively. The corresponding three-run spans are approximately 7.36–7.94 and 8.85–9.19. Both event times therefore remain within the digitized scatter, although the calculation misses the fastest interface rise. These comparisons support the non-geysering outcome and the bulk pressure and water-level scales, but they do not show that one baseline simulation recovers every spatial and temporal observable. The late terminal depressurization and remaining shape differences are consistent with unresolved exchange between the coherent pocket and the draining film. This sensitivity alone cannot identify a unique mechanism.

The narrow tower exposes a more localized error. The selected tracked-front calculation reproduces the pre-eruption stand ($0.834L$ versus $0.823L$), the pressure plateau, and the geysering sequence. Gas first enters the tower at $T^*_{\mathrm{rel}} = 3.622$ rather than the experimental 3.648–3.742. The one-cell grid-complete rim event occurs at 3.923, within the measured 3.900–3.981 span, and the displayed pressure falls below $H^* = 0.30$ at 4.097, within the 4.02–4.18 collapse envelope. The main residual is instead the early/high shock-fitted front (RMSE 0.206, bias +0.172), which catches the free surface at 4.051. Yet the connected cellwise $\alpha_g = 0.50$ field front reaches only $0.016L$.

The reported Y_{int} must therefore be interpreted as a reduced tracked-front closure, not a resolved dense-core isocontour. Diameter-group velocities reported in the source provide contextual scales rather than Case-B-specific error measures.

Campaign 2 reveals the consequence of this timing and exchange error for flow-regime classification. The model agrees with 5 of the 7 filmed Series-B outcomes and with the empirical criterion in 39 of the 63 simulated configurations. It recovers the geysering end member, the non-geysering regions with wide risers or small pockets, and the measured difference in pipe-end pressure response. All 24 criterion-based disagreements have the same sign: the criterion predicts geysering, whereas the model does not. This pattern is consistent with insufficient net water-column lift near the criterion boundary. It is not conservative for hazard screening because the model tends to miss geysering near that boundary rather than overpredict it.

Campaign 3 tests the role of the downstream boundary. Changing only this boundary reverses the outcome between A2 and B3, and the model captures that reversal. At the computed B3 peak pressure of 30.84 kPa, the experimental pressure–height regression gives approximately 2.50 m; the computed jet height is 2.17 m. The fixed pressure-wave celerity also underpredicts the short pressure peak (30.84 versus 55.03 kPa). For C9, the computed first pressure peak is 11.49 kPa at 0.54 s, compared with 10.69 kPa at 0.50 s. The predicted oscillation period is shorter than measured. Later pocket-driven geysers lie outside the composite one-dimensional representation. The chamber model therefore captures the initial transient response but not the later gas-pocket dynamics.

5.2. Capabilities of the reduced model

For the entrapped-air comparisons in Campaigns 1–2, the model determines the geysering outcome without a prescribed gas-release history. Pocket pressure evolves from the computed gas mass and volume; the coupled solution then determines gas transport into the shaft and water displacement. The response consequently changes with riser diameter and pocket volume. Campaign 3 provides a separate test of the reduced hydraulic branches through the downstream boundary and the initial all-water transient. In the entrapped-air cases, a liquid-only calculation could match a selected trace only by prescribing the gas-side forcing and therefore could not determine the outcome independently.

The comparisons also show which quantities are reproduced most consistently at one-dimensional resolution. Slowly varying pressure plateaus, quasi-static water-column levels, maximum free-surface elevations, and the first mass-oscillation peak agree more closely than interface breakthrough, annular-film motion, and short pressure peaks. Gas inventory, liquid-column inertia, and hydrostatic balance primarily control the former quantities. The latter depend more strongly on local three-dimensional exchange and unresolved time scales at the junction and shaft wall.

5.3. Closure limitations revealed by the experiments

The shaft-mouth closure remains the main limitation. In the narrow-tower configuration, the resolved driven regime activates a crown-referenced pressure datum and a pocket-fed gas core. It also activates conserved moving-front and slug dynamics, a glug resistance, and a finite mouth aperture. The

Case-B resistance and aperture pair was chosen in the disclosed bounded in-sample screen. Together, these elements form a regime-conditioned closure rather than a unified cross-sectional junction solution. The wide tower uses an invert-referenced datum, whereas the narrow tower requires the physical crown connection. A future junction model should transmit the vertical pressure distribution across the mouth instead of reducing it to a single scalar datum.

A second limitation concerns the coupling between cavity propagation and vent feed. The fixed cavity-front celerity places the initial arrivals close to the observed scatter, but the same reduced feed slows the pocket-fed core and delays breakthrough. Separating front speed from the pressure-driven gas flux through the mouth would allow the two constraints to be evaluated independently.

A third limitation is the volume-neutral counter-current exchange used at the tee. This exchange preserves pocket pressure during balanced venting and is needed to produce the B-H1 geyser. In reservoir-fed cases with wider risers, however, it cancels part of the water-column displacement caused by entering gas. Removing the exchange restores lift in those cases, but it also removes the B-H1 pressure rebuild. Both branches therefore require a flux-limited exchange law based on local counter-current capacity rather than a fixed equality of gas and liquid volumes.

Dimensional reduction introduces two further limits. The fixed Taylor-core closure does not resolve the evolving connected-pocket topology or the draining wall film in the wide tower. The reduced chamber and wave celerity do not reproduce the shortest B3 pressure peak or the later arrival and venting of pockets in C9. Addressing these discrepancies requires targeted closure development or local multidimensional coupling, not adjustment of a global parameter to individual tests.

5.4. Implications and scope of use

The present evidence supports using the model to examine mechanisms and system-scale scenarios away from the empirical criterion boundary. One computational description tracks gas inventory, pocket compression, the geysering outcome, and the bulk pressure and water-column response. The evidence does not support stand-alone safety classification near the boundary. Every Campaign 2 criterion disagreement is a missed geysering case, and the local B3 pressure peak remains sensitive to the pressure-wave representation.

The reported accuracy applies only to the tested configurations and the stated regime-conditioned junction closures. Visualization-only and sensitivity results retain the narrower roles defined in their captions. These comparisons do not establish applicability across shaft shapes, manhole covers, network interactions, or prototype Reynolds and Weber numbers. The next validation step should use an independent apparatus without changing the closure parameters. Simultaneous measurements of gas flux, liquid return flow, pocket pressure, and interface topology at the shaft mouth are the priority. Such data could distinguish an exchange-law error from a shaft-entrainment error and make the next closure revision falsifiable.

6. Conclusions

This study assessed case-specific one-dimensional realizations of a gas-conserving mixed-flow framework against three published experimental data sets. Horizontal conduits and vertical shafts were coupled with apparatus-appropriate reductions, and the gas-release history was computed rather than prescribed. For the two ventilation-shaft cases, the corresponding realizations selected the observed non-geysering and geysering outcomes. In the wide tower, the separately identified connected-pocket sensitivity recovered the pressure and maximum water-level scales. Its interface event times remained within the three-run spread, but terminal depressurization was late. In the narrow tower, the selected tracked-front calculation recovered the pre-eruption pressure and water-level scales. Grid-complete rim contact and pressure collapse fell within the measured intervals, although the reduced interface trajectory remained early and high.

For the horizontal-pipe/vertical-riser data set, the model agreed with 5 of 7 filmed outcomes and with the empirical criterion in 39 of 63 simulated configurations. In all 24 criterion-based disagreements, the criterion predicted geysering and the model did not. For the junction-chamber data set, the model recovered the A2/B3 change in geysering outcome and the first C9 pressure transient. It underpredicted the short B3 pressure peak and did not represent the later pocket-driven events.

For Campaigns 1–2, explicit gas conservation and transport recover the main geysering outcome patterns, bulk pressure levels, and water-column response. Campaign 3 adds evidence that the reduced hydraulic formulation captures the boundary-driven A2/B3 branch flip and the first C9 transient. The remaining errors are associated with shaft-mouth exchange, connected gas-core and wall-film dynamics, pressure-wave representation, and chamber reduction. These realizations are therefore suitable for mechanistic assessment away from the empirical criterion boundary. Safety-critical classification near that boundary requires a junction model that conserves phase fluxes across the full cross section and resolves entrainment.

Data Availability Statement

The data and materials supporting the findings of this study are available from the corresponding author upon reasonable request. A persistent archive identifier will be added if the project repository is deposited before submission.

Declaration of Generative AI Use

Generative AI tools were used to assist language revision and manuscript organization. The authors critically reviewed and edited all generated material and take full responsibility for the content of the manuscript.

References

[1] S. J. Wright, J. W. Lewis, J. G. Vasconcelos, Geysering in rapidly filling storm-water tunnels, Journal of Hydraulic Engineering 137 (1) (2011) 112–115. doi:10.1061/(ASCE)HY.1943-7900.0000245.

[2] J. G. Vasconcelos, S. J. Wright, Geysering generated by large air pockets released through water-filled ventilation shafts, Journal of Hydraulic Engineering 137 (5) (2011) 543–555. doi:10.1061/(ASCE)HY.1943-7900.0000332.

[3] J. Cong, S. N. Chan, J. H. W. Lee, Geyser formation by release of entrapped air from horizontal pipe into vertical shaft, Journal of Hydraulic Engineering 143 (9) (2017) 04017039. doi:10.1061/(ASCE)HY.1943-7900.0001332.

[4] K. Z. Muller, J. Wang, J. G. Vasconcelos, Water displacement in shafts and geysering created by uncontrolled air pocket releases, Journal of Hydraulic Engineering 143 (10) (2017) 04017043. doi:10.1061/(ASCE)HY.1943-7900.0001362.

[5] L. Liu, W. Shao, D. Z. Zhu, Experimental study on stormwater geyser in vertical shaft above junction chamber, Journal of Hydraulic Engineering 146 (2) (2020) 04019055. doi:10.1061/(ASCE)HY.1943-7900.0001660.

[6] Y. Zhang, Y. Chen, S. Qian, H. Xu, J. Feng, X. Wang, Experimental study on geysers in covered manholes during release of air pockets in stormwater systems, Journal of Hydraulic Engineering 148 (5) (2022) 06022003. doi:10.1061/(ASCE)HY.1943-7900.0001978.

[7] Y. Qian, D. Z. Zhu, L. Liu, W. Shao, S. Edwini-Bonsu, F. Zhou, Numerical and experimental study on mitigation of storm geysers in edmonton, alberta, canada, Journal of Hydraulic Engineering 146 (3) (2020) 04019069. doi:10.1061/(ASCE)HY.1943-7900.0001684.

[8] G. Zhou, A. Prosperetti, Violent expansion of a rising Taylor bubble, Physical Review Fluids 4 (7) (2019) 073903. doi:10.1103/PhysRevFluids.4.073903.

[9] S. N. Chan, J. Cong, J. H. W. Lee, 3D numerical modeling of geyser formation by release of entrapped air from horizontal pipe into vertical shaft, Journal of Hydraulic Engineering 144 (3) (2018) 04017071. doi:10.1061/(ASCE)HY.1943-7900.0001416.

[10] J. G. Vasconcelos, S. J. Wright, P. L. Roe, Improved simulation of flow regime transition in sewers: two-component pressure approach, Journal of Hydraulic Engineering 132 (6) (2006) 553–562. doi:10.1061/(ASCE)0733-9429(2006)132:6(553).

[11] Z. Xue, L. Zhou, D. Z. Zhu, A conservative cut-cell finite-volume framework for transient mixed flows with heterogeneous branch coupling, manuscript submitted concurrently (2026).

[12] J. H. Stuhmiller, The influence of interfacial pressure forces on the character of two-phase flow model equations, International Journal of Multiphase Flow 3 (6) (1977) 551–560. doi:10.1016/0301-9322(77)90029-5.

[13] R. Borges, M. Carmona, B. Costa, W. S. Don, An improved weighted essentially non-oscillatory scheme for hyperbolic conservation laws, Journal of Computational Physics 227 (6) (2008) 3191–3211. doi:10.1016/j.jcp.2007.11.038.

[14] B. van Leer, Towards the ultimate conservative difference scheme. V. a second-order sequel to Godunov's method, Journal of Computational Physics 32 (1) (1979) 101–136. doi:10.1016/0021-9991(79)90145-1.

[15] A. R. Winters, C. Czernik, M. B. Schily, G. J. Gassner, Entropy stable numerical approximations for the isothermal and polytropic Euler equations, BIT Numerical Mathematics 60 (2020) 791–824. doi:10.1007/s10543-019-00789-w.

[16] J. Chan, K. Shukla, X. Wu, R. Liu, P. Nalluri, High order entropy stable schemes for the quasi-one-dimensional shallow water and compressible Euler equations, Journal of Computational Physics 504 (2024) 112876. doi:10.1016/j.jcp.2024.112876.

[17] E. F. Toro, Shock-Capturing Methods for Free-Surface Shallow Flows, John Wiley & Sons, Chichester, 2001.

[18] S. Gottlieb, C.-W. Shu, E. Tadmor, Strong stability-preserving high-order time discretization methods, SIAM Review 43 (1) (2001) 89–112. doi:10.1137/S003614450036757X.

[19] A. S. León, I. S. Elayeb, Y. Tang, An experimental study on violent geysers in vertical pipes, Journal of Hydraulic Research 57 (3) (2019) 283–294. doi:10.1080/00221686.2018.1494052.